Ontology of Transition

Causal Order, External Time, and the Thermodynamics of Physical Clock Records

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NotebookLM

Abstract

This volume develops Ontology of Transition, a framework in which transition rather than the static thing is the primitive unit of description. The underlying structure is a locally finite partial order of events, with no continuous physical time assumed. Time is represented in two operationally distinct ways: a system's internal time accumulates as a functional of its own changes, whereas its accessible external time is reconstructed from a physical record of a selected clock process delivered through a communication channel. Because the channel admits noise, delay, loss, and aggregation, external time is relative, local, and partially observable; the continuous parameter t enters only as calibration and a refinement limit, not as a primitive. Thermodynamic balances are defined on an invariant energy skeleton of the history, making them independent of the arbitrary linearization of concurrent events, and a Separation Principle organizes time, work, entropy production, and information-theoretic distinguishability as distinct functionals of one causal carrier. Part I develops the formal framework and proves its invariance backbone (Proposition I.1 and Corollary I.1); Parts II and III supply the reconstruction, identifiability, and thermodynamic theorems enabled by that framework.

Part II proves the reconstruction and identifiability limits. For the erasure channel, the minimax error of reconstructing the source coordinate is exactly n p σ^2, with perfect recoverability iff calibration is deterministic, and an additional counting floor when the tick count is unknown. Centrally, for the aggregating channel the pair (loss rate, calibration law) is never identifiable from the record law for any positive loss: it is determined only up to a one-parameter semigroup orbit of geometric compounding, containing at most one geometrically indecomposable representative — the rate of external time is gauge, and provenance metadata collapse the orbit. When the register output is a time-antisymmetric net current, it obeys a thermodynamic-uncertainty precision floor; objecthood also acquires a sharp critical loss rate p_c = $(\delta+\varepsilon)/(1-\delta-\varepsilon)$, set by the recognition scheme's tolerances δ and ε, above which fast object tokens dissolve while frozen ones persist at any loss.

Part III quantifies the work required for reuse. For a finite classical footprint reset isothermally without accessible side information, the average renewal work is bounded below by k_B T times the indirect rate–distortion function of the physical delivery channel — an indirect rate–distortion problem in the sense of Dobrushin–Tsybakov, Wolf–Ziv, and Witsenhausen. Closed forms are derived and shown exactly attainable; target distortions below the channel's Bayes risk are unattainable at any reset budget; and a sequential-ring corollary shows that renewal pays for the entropy of unpredictable innovations, not for the number of ticks. A three-floor synthesis proves the channel, register, and renewal budgets mutually irreducible: they constrain different observables through different mechanisms, and none implies another.

A recognition scheme operationalizes thinghood: physical objects appear as stable macro-regimes (types) and individual realizations (tokens) that persist against the flow of transitions, with a sharp phase boundary for that persistence derived in Part II. The volume connects the event-ordering tradition of distributed systems, rate–distortion theory with its indirect branch, and stochastic thermodynamics in one formal language. All quantitative results are checked numerically; the closed forms agree with a Blahut–Arimoto solver to machine precision.

Keywords: *process ontology; causal order; event time; external clocks; discrete-event systems; stochastic thermodynamics; entropy production; indirect rate–distortion; Landauer principle; gauge freedom of external time; object type; object token*

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ONTOLOGY OF TRANSITION—PART I

Causal Order of Events, Internal and External Clocks, Thermodynamics, and Information-Theoretic Distinguishability

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Abstract

This article is Part I of the three-part scientific work *Ontology of Transition.* The work's central idea is that transition, rather than the static thing, is the primary unit of physical and ontological description, and that operational time arises from physically recorded and communicated transitions rather than existing as an independently accessible primitive.

This article takes transition rather than thing as the primary unit of ontological description. Its primitive structure is a locally finite causally ordered event set without presupposed numerical time. A system S is embedded in a metasystem M. The reading of S's internal event clock is an additive functional of selected intrinsic transitions; its accessible external-clock coordinate is a functional of a physical record of CM outside S. Transmission and registration may omit, aggregate, delay, or distort ticks, making the record partial and local. Calibration assigns durations to discrete ticks; continuous time enters only by interpolation or a scaling limit. Energy and calibrated skeletons make thermodynamic and path functionals invariant under admissible linearizations. A separation principle states that time, work, information-theoretic distinguishability, and entropy production share transition histories but are not identical: they differ in arguments, dependencies, symmetry properties, and physical prerequisites. Equilibrium, quasistatic, cyclic, and protocol-dependent counterexamples establish their irreducibility; their relations are conditional balances and bounds. Observation is a discrimination channel whose cost belongs to its physical implementation. A recognition scheme separates ensemble-stable object types from realization-stable object tokens. A thing is a stable, recognizable organization of transitions; operational clock discreteness does not imply discrete spacetime.

Keywords: process ontology; causal order; event time; external clocks; partial observability; discrete-event system; stochastic thermodynamics; entropy production; KL divergence; object type; object token.

From Thing to Transition

The conventional descriptive scheme presupposes a thing prior to its change:

$$\text{thing} \longrightarrow \text{change of the thing} \tag{1}$$

First an object is identified, then a state is ascribed to it, and only afterward is a change of that state considered. This sequence is natural in language and everyday experience, but it is not obligatory as an ontological commitment. The present work adopts the reverse order:

$$\text{transitions} \longrightarrow \text{stable regimes of transitions} \longrightarrow \text{things} \tag{2}$$

The thing is not thereby denied. Rather, it receives a derivative definition: a stable fragment of a process that preserves recognizable features over a certain time scale. An object is not the absence of change but a reproducible organization of change.

The notation

$$S_0 \longrightarrow S_1 \tag{3}$$

records only the endpoints. It does not indicate how long the transition lasted, which path realized it, how much work was performed, whether irreversibility arose, or whether the selected observation channel can distinguish the forward history from its time reverse. The basic unit of description below is therefore an explicit dynamical specification of a transition.

Contribution of This Work

The physical relations used in §§ 6–10 are established results of stochastic thermodynamics and information theory. The decomposition of energy into work and heat, the trajectory-level expression for entropy production, the data-processing inequality, nonequilibrium free energy, and Landauer's principle are not presented here as new results [21, 8, 29, 28, 24].

The article's original contribution consists of five steps:

1. the precedence relation among events is adopted as the primitive structure of description, while scalar clocks are constructed only on causally ordered local histories;

2. a system's internal time is defined by the accumulation of its own transitions, whereas its external time is defined by a physically transmitted record of a selected clock process in the metasystem;

3. partial observability of external states is introduced explicitly through a channel that may conceal events and thereby generate local time coordinates that are generally asynchronous;

4. continuous time parameterization and thermodynamic and informational functionals are separated from causal order, while energy balances are defined on an invariant energy skeleton of the history;

5. objecthood is operationalized through a recognition scheme in which ensemble stability and the stability of an individual realization are distinguished as object type and object token, respectively.

Thus, the article proposes neither a new formula of stochastic thermodynamics nor a proof of the fundamental discreteness of spacetime. It develops an original formal framework in which causal order, local clock records, thermodynamic and informational functionals, and operational criteria of objecthood are jointly defined, while their non-equivalence and invariance properties are established under explicit conditions. A worked example exercising the clock channel and the recognition scheme end to end is given in Appendix B.

The principal notation of the framework is collected below; each symbol is defined at the indicated equation or section and used consistently thereafter.

Symbol	Meaning	Defined at
EM=(EM, ≺M, ℓM)	labeled, locally finite causal partial order of the metasystem	(4)–(6)
H; n	causally closed configuration; local chain of a sequential subsystem	(7)–(8)
xk, k, pkS	system state, protocol value, and system marginal on the k-th slice	(11)
ev; th	stochastic transition specification; its thermodynamic extension	(12); (40)
Sint, S,wint	internal event time of the system (weighted count of intrinsic changes)	(15)–(17)
C; Cint; tC	clock process; its internal tick count; its calibrated source coordinate	(19); (26)
ρ; rj; wj	delivery record; register state; decoded tick increment	(20), (A1)
S|Cext; tS|C	external time accessible to the system; its decoded calibrated coordinate	(20); (27)
KCS,; OS, OM	history-dependent clock channel; state-observation channels	§ 3.2; (32)–(36)
OD; CS,O,pathD, CS,O,ensD	typed map of admissible characteristics; record of characteristics	(37)–(39)
skE; skclk	energy skeleton; calibrated skeleton of an execution	§ 6, Prop. 1
σ; Σ; DOSend; AOS	pathwise and mean entropy production; endpoint and path distinguishability	(53), (56); (57)–(58)
R*; Ba; ε; h	recognition scheme; type region; tolerance; averaging window	(74)
ℓint, ℓext	internal and external durations of an object segment	(82)

Specification of a Transition

Let M be a metasystem containing the system under study S, and let

$$EM = EM, \prec M, \ell M \tag{4}$$

be its labeled, locally finite causal partial order. Here EM is the set of events, ℓM their labels, and ≺M a strict precedence relation:

$$e \prec Me, \qquad e \prec Mff \prec Mge \prec Mg \tag{5}$$

The relation ≺M is taken as a primitive of the construction: e≺Mf means that the outcome of e may enter the conditions of f. It is not defined in terms of a numerical instant of time. For comparable events, local finiteness is additionally assumed:

$$\#\{gEM : e \prec Mg \prec Mf\} < \qquad e \prec Mf \tag{6}$$

Incomparable events may occur in parallel. Event conflict is not modeled separately in the present construction. Consequently, the global history of M is generally partially rather than linearly ordered. Scalar clocks do not reconstruct this entire structure: they provide an order-preserving valuation on a selected causal chain; a total order requires an additional rule for incomparable events. This separation between event order and clock readings accords with the logic of happened-before in distributed systems and employs the causal layer incorporated into the richer formalism of event structures [20, 32].

A realized configuration HEM is assumed to be causally closed:

$$eH,\, f \prec MefH \tag{7}$$

The local history of a single sequential subsystem is a chain

$$n = e1 \prec Me2 \prec M \prec Men \tag{8}$$

To associate events with states of the metasystem, one selects a nested sequence of causally closed configurations

$$H0H1Hn, \qquad Hk = Hk - 1\{ek\}, \qquad Hk = Hk \tag{9}$$

This selected linear execution does not render incomparable events causally related.

When the specification includes a control protocol, both the system state and the protocol value are likewise defined as functions of a causally closed configuration: x=xH and =H. Let ES denote events that change x, and E events that change , under an admissible addition of an event to a configuration. For the thermodynamic history under consideration, each of the sets ES and E forms a local chain. If a single physical event changes both x and , it is refined into two causally ordered elementary substeps; hence these sets are taken to be disjoint below. We adopt the axiom of protocol–system comparability:

$$uE\,vES : \qquad u \prec Mv\ v \prec Mu \tag{10}$$

It follows that the thermodynamically significant set Eth=ESE is itself a chain and has the same order in every linear execution of the given causal order. An event outside Eth is called thermodynamically silent if its admissible addition does not change the pair x,. On the selected slices, states and distributions take the form

$$mk = mHk, \qquad xk = Smk, \qquad k = Hk, \qquad pkS = S\#pkM \tag{11}$$

Here S:XMXS specifies the boundary and state space of the system S. The phrase “state change” means a difference between configurations on two nested slices; it does not presuppose an already introduced continuous parameter.

A general stochastic specification of a transition is given by the tuple

$$ev = EM, XM, S, , p0M, \bullet, PM, \tag{12}$$

where p0MPXM is the initial distribution; •=H0,,Hn takes values in and represents a single configuration-dependent protocol variable on the selected slices; and PM is a measure on causally admissible histories. This notation also encompasses non-Markovian models. In the fixed-length n Markovian specialization, the measure is generated by transition kernels Kk:

$$pk + 1M = KkpkM, \qquad P, nMdm0, , dmn = p0Mdm0k = 0n - 1Kkdmk + 1 \mid mk \tag{13}$$

For histories of random length, a stopping law is specified in addition. System trajectory measures are obtained as pushforward measures under the deterministic projection:

$$PS = S, \#PM \tag{14}$$

Here pkM describes the ensemble of the metasystem on the kth slice, pkS is its system marginal, and =m0,e1,m1,,en,mn is an individual history. The same initial and final configurations may be connected by different event histories; consequently, clocks, work, entropy production, and path distinguishability are generally not determined by endpoints alone.

Local Clocks: Internal and External Time

Internal Event Time of a System

Let n be a local history of S, and let xk=Smk be its states on successive causal slices. The predicate of an intrinsic change is

$$Sek = 1\{xkxk - 1\} \tag{15}$$

For a nonnegative weight function wS:XSXSR0 that is positive for every included change xx′, internal event time is defined as an additive measure of changes along the local history:

$$S, wintn = k = 1nwSxk - 1, xkSek \tag{16}$$

With unit weights, each intrinsic transition generates one internal tick:

$$Sintn = k = 1nSek = NSn \tag{17}$$

Internal time is not a medium in which events occur: its reading is changed by the events themselves. Nor does it generate their order—the order is already specified by the relation ≺M. This breaks the possible logical circle in which "change is defined in time, while time is defined through change."

In the theory of continuous-time random walks, the jump number is called discrete operational time [15]. Here the expression "internal event time" is used as the author's term for a measure of a system's own changes and does not claim the conventional meaning of internal time in other fields.

The ensemble-averaged internal time after n events in the local record is

$$Sintn = EPSSintn \tag{18}$$

With unit weights it measures mean dynamical activity, and for general wS it measures mean weighted activity, but not entropy, irreversibility, or work. A reversible transition also generates an internal tick; a spontaneous fluctuation may change the state without work by an external protocol. The number of ticks depends on the boundary of S, its state space, and the model's resolution. For smooth continuous trajectories, the counter must be replaced by an additional metric, path variation, or another clock function, and such a construction is no longer canonical.

External Clocks as a Metasystem Process

Because the global history of M may contain parallel events, merely counting all changes in the metasystem does not define a unique scalar clock. Within the selected decomposition, let us identify a clock subsystem C such that CM and CS=∅. It lies outside the boundary of S and has a causally ordered chain of ticks

$$C = c1 \prec Mc2 \prec M, \qquad Cintcn = k = 1nwCck, \quad wCck > 0 \tag{19}$$

The process C is internal to the metasystem and serves as an external reference only relative to the boundary of S. If S significantly affects the order or statistics of the ticks of C, the joint dynamics of S+C must be modeled explicitly.

For C to serve as an external clock for S, its history must physically produce a record =1,2, accessible to the system through a history-dependent channel KCS,. External time is a functional of this record rather than a reading of the entire metasystem. Let rj be the register state after the jth attempt, and let wj0 be the decoded increment. When rjrj−1, the external time accessible to the system is

$$S \mid Cextn = j = 1nwj\,1\{rjrj - 1\} \tag{20}$$

The index j denotes the index of the source tick that generated the jth genuine, nonaggregated attempt; the complete message-provenance model is given in Appendix A. In an ideal channel without loss, duplication, or errors, j=j, wj=wCcj, and S|Cext=Cint at the corresponding events. In general, there is no literal equality. Let C, be the measurable projection of the complete history of M onto the clock history of C. Then

$$PCclock = C, \#PM \tag{21}$$

At the level of probability laws on histories, the channel induces the distribution of output records

$$PS \mid CrecB = \int KCS, B \mid C\, PCclockdCKCS, *PCclock \tag{22}$$

The reading (20) is a functional Ftick of a particular output record. For a stochastic, noisy, or incomplete channel, the appropriate state of knowledge of S may be a conditional distribution of the hidden source reading. Thus, the external states of the metasystem are not assumed to be fully observable to S: the channel may omit, aggregate, delay, or distort ticks. For a deterministic channel that neither creates spurious events nor duplicates them,

$$0NS \mid CextHNCintH \tag{23}$$

Here NS|Cext is the number of successful registrations, and both quantities are counted within the same causally bounded window H. Noise may generate false ticks, in which case (23) must be replaced by an error model. Memory updating, decoding, the filtering distribution of the hidden clock state, and conditions for recovering message order are deferred to Appendix A.

Registration, Relativity, and Synchronization

An external tick is operationally accessible to the system only insofar as it leaves a distinguishable trace. A successful attempt j becomes a registration event when

$$x, rj - 1jx, rj, \qquad rjrj - 1 \tag{24}$$

Reading an external clock is therefore itself an internal transition of the extended system S+=SR. This does not erase the distinction between the two times: Sint pertains to changes of the selected core S, whereas S|Cext pertains to changes in the register R causally induced by the reference C. The system boundary must be specified explicitly.

On a single segment of a history, one may have

$$Sint = 0, \qquad S \mid Cext > 0, \tag{25}$$

when the core S does not change while the external register receives ticks. The converse is also possible: several intrinsic transitions of S may occur between two registered ticks of C. Hence internal and external time cannot be reduced to a single counter.

Systems S1 and S2 that use different clock processes or channels generally acquire different external coordinates. Synchronizing them requires either shared reference events or a protocol for exchanging records. Logical synchronization may reconcile the order of causally related events, but by itself it does not establish equal rates, equal durations, or physical simultaneity.

Calibration and Continuous Representation

Calibration of the source assigns positive increments of physical duration to the ticks of C:

$$tCcn = t * +k = 1nk, \qquad k > 0 \tag{26}$$

After message loss or aggregation, the system has its own decoded coordinate. Here the index ℓ refers to a delivery attempt rather than directly to a source tick: the duration decoder assigns a nonnegative increment ℓ. Throughout, an increment indexed by a source tick is a source quantity, while an increment indexed by a delivery attempt is its decoded counterpart. For a genuine, nonaggregated signal under exact decoding, ℓ=ℓ; in an ideal channel, ℓ=ℓ and ℓ=ℓ>0. Aggregated and spurious signals are treated in Appendix A. The decoded coordinate is

$$tS \mid Cj = t * +\ell = 1j\ell 1\{r\ell r\ell - 1\}, \qquad \ell 0 \tag{27}$$

In general, tS|Cj does not coincide with tCcj. Even when their values belong to R, the domains of both coordinates remain discrete. A continuous coordinate is introduced through interpolation or obtained within a separately specified family of increasingly fine clocks whose mesh tends to zero. Causal order alone specifies "earlier/later" where events are comparable; the choice of a reference, unit, and measure adds "by how much."

For Markovian dynamics directly on XS, a continuous generator appears as the limiting representation of the full transition kernel on the clock grid, including the probability of no jump:

$$Ptk, tk + 1S = I + tkLk + otk, \qquad ktk0, \tag{28}$$

from which, under the appropriate convergence conditions, it follows that

$$tptS = LtptS \tag{29}$$

For a Markov jump process directly on XS, when all state transitions are counted, the rate of mean weighted internal time is

$$dS, wintdt = xptSxx'xkxx'twSx, x' \tag{30}$$

For wS1, this is the usual dynamical activity. Thus, stationarity of the distribution is compatible with continuing internal transitions:

$$tptS = 0, \qquad dSintdt > 0 \tag{31}$$

The relation between the number of configuration changes and dynamical activity is used in the statistical mechanics of trajectories [22, 30]. All subsequent continuous-time notation refers to such a calibrated clock representation and does not reinstate t as an ontological primitive.

The hierarchy has four levels: events define causal order; internal clocks accumulate a system's own changes; the external channel transmits to the system a record of a selected clock process; and calibration assigns duration to ticks.

Observation as a Channel of Discrimination

The clock channel KCS, is a history-dependent observation channel. A memoryless state-observation channel is a simpler special case. In the deterministic formulation for the metasystem,

$$OM : XMY, \tag{32}$$

whereas in the general case it is given by a Markov kernel

$$OMdy \mid m \tag{33}$$

The channel determines which distinctions in XM are preserved in the observable space Y. For deterministic OM, states satisfying

$$mOMm' \qquad OMm = OMm' \tag{34}$$

are indistinguishable. A transition within a single equivalence class is a hidden event. Several hidden transitions, including a cycle that returns to the previous observable value, may leave no tick in the record. For a general kernel, its action on a distribution is defined as

$$pkOMB = XMOMB \mid m\, pkMdmOM * pkM \tag{35}$$

In the deterministic case, OM*p=OM#p. Correlated noise and observational memory are represented by a separate kernel OM,dy•| on the space of histories. The observed trajectory measure is

$$POMB = \int OM, B \mid \; PMdOM, *PM \tag{36}$$

Even if the complete dynamics of M is Markovian, the coarse-grained dynamics in Y may have memory: hidden degrees of freedom carry information between observations. The Markov property of the observed process therefore cannot be assumed automatically.

For observations of the system alone, the channel OS:XS⇝Y is used below. If access to M occurs exclusively through S, then in the deterministic case OM=OSS; kernels are composed accordingly. Neither OM nor OS requires a conscious observer. Either may be a measuring instrument, a receptor, a classification algorithm, or a statistical partition of the state space. Thermodynamic cost attaches not to an abstract channel as such but to its concrete physical realization (§ 11).

An Extensible Map of Characteristics

Not every system has an energy function, a thermal environment, and a notion of heat. A unified framework must therefore not be turned into a requirement to ascribe W, Q, and to a text, an institution, or an arbitrary computational description.

Let D be a class of admissible specifications for which the available structures have been stated in advance. For each transition, define the set of admissible characteristics and the corresponding typed map:

$$AD = \{: FisdefinedintheclassD\}, \qquad OD = (F)AD \tag{37}$$

A component enters the map only when its necessary prerequisites are specified in the model. At the level of an individual realization and at the ensemble level, respectively, this yields two mutually consistent expressions:

$$CS, O, pathD, = EM; Sint, S \mid Cext; O, pathD,, \tag{38}$$

$$CS, O, ensD = EM; Sint, S \mid Cext; OD \tag{39}$$

The first semicolon separates the primitive causal order of events from the local clocks; the second separates the clocks from the remaining derivative characteristics. The calibrated coordinate tC is added to the map only for a class D in which a reference, a unit, and an interpolation rule are specified. External clocks are not a complete measure of the activity of M: they pertain to the triple S,C,KCS,.

For a stochastic system, changes in distributional entropy and informational distinguishabilities may be defined. For continuous X, one must fix a base measure or use relative entropy: differential entropy itself depends on the coordinates. After a calibrated clock representation has been selected, a thermodynamic specification takes the form

$$th = ev, tC, E, Tb, Ldb, \tag{40}$$

where Ex, is the energy function, Tb is the temperature of the thermal environment, and Ldb denotes the conditions that make the dynamics consistent with the energetics, in particular local detailed balance. Energetics and dynamics play different roles and must be connected by the physical model. For such a specification, the map extends to

$$OStheta = \langle W\rangle th, \langle Q\rangle th, Ssysth, \langle\rangle th, DOSendth, AOSth \tag{41}$$

Here ⟨⟩th denotes mean trajectory-level entropy production; its definition through the forward and reverse process measures and its relation to are given in § 7. This notation asserts neither equality nor interchangeability among the components. The components are constructed from one process specification but have different mathematical types and levels of description.

The Separation Principle: Why the Projections of Transition Are Not Identical

A transition history provides common input to several characteristics, but is itself neither a clock reading, work, information-theoretic distinguishability, nor entropy production. Inferring an equality of time and work from the fact that both rely on changes of state commits the fallacy of the undistributed middle. Sharing a process specification does not make the resulting maps equal: clock readings, work, information-theoretic distinguishability, and entropy production have different domains, dependencies, and transformation properties. We refer to this claim as the separation principle.

Let zk=xk,k,ck,rk, be an extended state containing the system state, protocol, selected clock, and accessible record, and let =z0z1 be a history in a path space . The relevant valuations have schematically different arguments:

$$;S, wS \mapsto S, wintR0, ; E, \mapsto WR, P, Q; O \mapsto DOP, Q = DKLO * Q0, , PF, PR \mapsto = DKLPR0,$$

The last identification has the physical meaning of entropy production only under the conditions stated in § 7. External time has a further type: it is a functional of a record generated by a selected clock process and transmission channel, rather than a functional of the complete metasystem history.

Causal order is fixed by ≺M; a clock neither creates nor replaces that order. Internal event time counts or weights selected intrinsic transitions at a stated system boundary and resolution. External time assigns readings to registered changes of a reference process. Both constructions are chronometric and depend on a clock scheme; neither is determined by energy transfer.

Work is an energetic valuation. Under convention (43), a protocol update kk+1 at fixed xk contributes

$$Wk = Exk, k + 1 - Exk, k,$$

whereas a state transition at fixed protocol may contribute heat while producing no work. Thus every work-bearing act can be represented as a transition of an extended state, but not every transition is work. The bare sequence x0,x1, is insufficient to determine W; the energetic structure and the work channel must also be specified.

Information-theoretic distinguishability is relational rather than a count of events. It compares two specified probability measures after a specified observation channel has acted on them. A physical record requires a distinguishable change or correlation in a receiver or memory, but that change realizes the record; it is not itself the value of the information measure. Many microscopic transitions may yield zero distinguishability at the selected observational level, while the absence of a registered event can be informative relative to a specified clock, observation window, and alternative measure.

Entropy production is distinct from both time and information-theoretic distinguishability in general. Under the hypotheses of § 7, it is the mean log-likelihood ratio between the forward path measure and the appropriately reversed path measure. Without those hypotheses, the same KL expression measures statistical irreversibility but need not represent thermodynamic entropy production.

The separations are exhibited by four counterexamples, C1–C4, collected in the table below. Information-theoretic zeros in the table refer only to the specified distinguishabilities, not to the absence of every kind of information in the system.

Model or history	**Internal time / activity**	**Work**	**Specified distinguishability**	**Mean entropy production**
C1. Stationary equilibrium jump dynamics with nonzero mean activity, at fixed and under detailed balance	The tick count is random, but its mean is positive	W=0	Dend=0; AO=0 if the observed forward and reversed measures coincide	=0
C2. Reversible quasistatic protocol between equilibria with F0	Depends on duration and resolution	⟨W⟩F0	Endpoint distinguishability may be nonzero if the selected channel distinguishes the initial and final equilibrium distributions; directional distinguishability tends to zero	0
C3. Nontrivial reversible loop returning to its initial state or distribution	Positive; for example, two ticks in xx′x	May be zero	Dend=0; directional distinguishability vanishes only for reversal-invariant measures	Zero under reversible conditions
C4. The same chain x0,,xn under two admissible protocol histories	The same under one internal-clock scheme	The histories are chosen so that W1W2	May differ together with the measures and observed records	May differ

Counterexample C1 rules out identities of time with work or entropy production. C2 separates reversible work from irreversibility and shows that event count does not determine work. C3 shows that accumulated time cannot be recovered from endpoint distinguishability. C4 explains why work is a functional of the extended history x, rather than of intrinsic state changes alone.

There is also a structural distinction under time reversal. With unit weights, or with weights satisfying wS(ϑx′,ϑx)=wS(x,x′), where ϑ is the state reversal of § 7, the intrinsic event measure is time-reversal even:

S,wint=S,wint.

Where the forward and reversed path measures are mutually absolutely continuous, trajectory-level entropy production is constructed from an oriented likelihood ratio and changes sign under the conjugate reversal:

=−.

Here is the analogous log-likelihood functional of the conjugate reverse process. In Markov jump models, the mean rate of the unit-weight event counter is the dynamical activity. The counter should not, however, be identified without further assumptions with the entire time-symmetric part of the path action, which may also contain waiting-time, escape-rate, and other kinetic contributions. Internal event time and entropy production therefore probe different sectors of the same stochastic history: the activity counter is even under path reversal, whereas the trajectory-level log-likelihood ratio is odd under conjugate reversal. Its mean is nonnegative and should not itself be described as an odd observable. Without an additionally specified inner product, these sectors should not be called orthogonal in the literal mathematical sense.

The projections are nevertheless physically related. The first law couples work and heat; Landauer inequalities constrain the cost of specified information-processing operations; thermodynamic speed limits jointly constrain duration, activity, and entropy production; and the data-processing inequality bounds observable distinguishability. Thermodynamic uncertainty relations connect the precision of steady-state currents with dissipation, while specific models of Brownian and autonomous clocks yield model-dependent trade-offs among accuracy, resolution, and cost whose form depends on the clock architecture and drive [2, 3, 12]. These are conditional balances and bounds, not universal identities or fixed conversion rates.

Thus, a transition history provides a common process specification; a clock reading is a chronometric functional, work an energetic path functional, information-theoretic distinguishability a specified statistical relation, and entropy production a thermodynamic measure of the distinguishability between forward and physically reversed path measures under the conditions of § 7.

Work, Heat, and State Entropy

For brevity, §§ 6–10 assume a finite or countable state space XS; in the continuous case, sums are replaced by integrals with respect to a specified reference measure. A calibrated clock record tk is fixed: it may be tCck under ideal access or the reconstructed coordinate tS|C. Below, ptS, t, and integrals over t denote an interpolation or the continuous limit of this discrete sequence. At the level of an individual realization, work, heat, and entropy production are denoted by W, Q, and ; angle brackets denote ensemble averages.

Proposition I.1 (skeleton invariance). Assume axiom (10), the local-chain structure of the sets of system and protocol events with the substep refinement of § 2, and a calibrated clock record assigned to events independently of the linear extension. Then (i) all causally admissible linear executions of a finite configuration share the same energy skeleton, and work, heat, and the change of system entropy factor through it; and (ii) all such executions share the same calibrated skeleton.

Proof. Let L and L′ be two causally admissible linear executions of the same finite configuration H. Since EthH is a chain, their restrictions to thermodynamically significant events coincide:

L|EthH=L′|EthH.

Let skEL denote the energy skeleton of an execution: its restriction to Eth after removing repetitions of the pair x, introduced by thermodynamically silent events. Then skEL=skEL′. Permuting the remaining causally incomparable events can only insert zero energy increments and does not change W, Q, Ssys, or other functionals that factor through this skeleton.

For time-resolved path measures, a calibrated clock record t=t0,t1, is additionally fixed on this local chain independently of the enumeration order of events outside Eth. The calibrated skeleton skclkL,t consists of the sequence x„t and retains grid nodes with a repeated pair x, and, in the continuous description, waiting times. Hence

skclkL,t=skclkL′,t,

but, unlike skE, this skeleton does not discard information about the absence of a jump between successive clock readings; this establishes the skeleton equalities. □

Corollary I.1 (invariance of derived functionals). Pathwise entropy production, its mean, and observed path distinguishability — functionals of the forward and reverse path measures constructed in § 7 by pushforward through the calibrated skeleton, with observed path distinguishability defined in § 8 — do not depend on the representative of the linearization class. The proof is immediate from Proposition I.1(ii).

Remark. Corollary I.1 becomes fully explicit in §§ 7–8: § 7 constructs the forward and reverse path measures by pushforward through the calibrated skeleton, while § 8 defines observed path distinguishability from their images under the observation channel. In §§ 6–10, the index k refers to the corresponding energy or calibrated skeleton, refined where necessary into explicitly ordered elementary substeps. If the assignment of the clock record to events itself depends on an arbitrary linear extension, invariance of time-resolved functionals is not claimed: a separate synchronization model is required. Functionals specifically observing events outside the indicated skeletons are likewise not covered by this claim.

At slice k of the selected clock, the mean energy of the system is

$$Uk = xXSpkSxEx, k \tag{42}$$

Under the discrete convention "protocol update first, state transition second," trajectory-level increments are defined by

$$Wk = Exk, k+1 - Exk, k, Qk = Exk+1, k+1 - Exk, k+1, Exk+1, k+1 - Exk, k = Wk + Qk \tag{43}$$

Their ensemble averages are

$$\langle W \rangle k = xpkSxEx, k+1 - Ex, k, \tag{44}$$

$$\langle Q \rangle k = xpk+1Sx - pkSxEx, k+1, \tag{45}$$

so that, under the sign convention Q>0 for heat absorbed by the system,

$$Uk+1 - Uk = \langle W \rangle k + \langle Q \rangle k \tag{46}$$

A different order of elementary substeps yields a different discretization but the same smooth limit. In this limit,

$$dU = \langle W \rangle + \langle Q \rangle, \qquad \langle W \rangle = xptSxEx, tdt, \qquad \langle Q \rangle = xEx, t\, dptSx \tag{47}$$

Here all work is assumed to be performed through changes in t. Nonconservative forces, chemical work, and particle exchange would require additional terms.

The integrated work of the external protocol is

$$\langle W \rangle th = t0t1xptSxEx, tt\, dt \tag{48}$$

This separation of work and heat is standard in stochastic energetics [29]. Work characterizes energy exchange induced by a change in the external protocol; it may either be performed on the system or extracted from it.

The entropy of a distribution is defined as

$$SptS = -kBxptSxlnptSx, \tag{49}$$

and its change

$$Ssys := Spt1S - Spt0S \tag{50}$$

depends only on the endpoint distributions. This quantity does not measure the total irreversibility of the process.

Entropy Production and Time Reversal

For an isothermal process at temperature Tb, the mean dimensionless total entropy production is

$$th = SsyskB - \langle Q \rangle thkBTb0 \tag{51}$$

To relate this quantity to path distinguishability, the conditions must be stated explicitly. In what follows, we assume:

1. Markovian stochastic dynamics;
2. local detailed balance with respect to E and Tb;
3. a physically correct reverse protocol;
4. reversal of all time-odd variables;
5. an initial distribution of the reverse process equal to the time-reversed final distribution of the forward process;
6. absolute continuity of the forward measure with respect to the reversed measure on the path set under consideration;
7. thermodynamic completeness of the state space XS: hidden variables carry no unaccounted energy or entropy fluxes.

Let reverse the odd state variables, and let reverse the odd control parameters. An element of the space of calibrated paths has the form

$$clk = x0, t0, 0, e1, x1, t1, 1, , en, xn, tn, n$$

The reversed grid, protocol, and initial distribution of the reverse process are defined by

$$tk = t0 + tn - tn - k, \qquad k = n - k, \qquad p0S = \#pnS$$

Set k=xn−k,tk,k. The time-reversed trajectory then has the compact form

$$clk = 0, en, 1, , e1, n \tag{52}$$

Here the protocol is indexed by the slices k=0„n. We assume the involutivity 2=id, e=e, and measurability of , so that 2=id on admissible calibrated trajectories. For even control parameters, =id. Merely reversing the order relation EMop does not yet define a physical reverse process: admissible reverse events, the protocol, and a separate measure must be specified. For a finite discrete step, the reverse measure also changes the order of the elementary substeps: the forward order "work, then heat" corresponds to "reverse heat, then reverse work." If the same order of substeps is retained, the thermodynamic identification below is claimed only in the smooth limit.

Let PthM,clk be the full path measure on calibrated causal histories. The path measure used below is its pushforward under the calibrated-skeleton map,

Pthclk=skclk#PthM,clk,

and includes nodes without a state change on the discrete grid or waiting times in the continuous description. It need not coincide with measure (13), defined on a selected sequence of configurations, and a fortiori does not coincide with the measure of the embedded chain if the latter is constructed conditional on the occurrence of a jump. The reverse measure is constructed in the same way on the time-reversed calibrated skeleton. Denote PF=Pthclk and PR=#Pthclk. When the clock record is fixed independently of linearization, , its mean, and the observed path distinguishability likewise do not depend on the representative of the linearization class; this proves Corollary I.1. For a realization clk, pathwise entropy production is defined by the Radon–Nikodym derivative

$$clk = lndPFdPRclk \tag{53}$$

The pathwise change in system entropy is

$$ssysclk = -kBlnpnSxn + kBlnp0Sx0 \tag{54}$$

Local detailed balance and the chosen physical time reversal yield the identification

$$clk = ssysclkkB - QclkkBTb \tag{55}$$

Without these physical conditions, formula (53) defines the statistical irreversibility of a trajectory, but not necessarily its thermodynamic entropy production.

Then

$$th = \langle\rangle PF = DKLPR \tag{56}$$

Relation (56) is a well-known result of stochastic thermodynamics [8, 28]. If S is a reduced observable description of a thermodynamically more complete system, the KL distinguishability of its paths generally provides only a lower bound on the full . Under a different choice of reverse or reference measure, the KL divergence remains a measure of the distinguishability of histories but need no longer coincide with physical entropy production. The result of Kawai, Parrondo, and Van den Broeck [17] concerns a related but more specific connection between dissipation and the distinguishability of phase-space distributions of the forward and reverse processes, and is not used here as a source for the general pathwise equality.

Trajectory-level time asymmetry can also be used to estimate dissipation from stationary records [26].

Endpoint Distinguishability and Distinguishability of the Direction of Time

The distinguishability of the final and initial observed distributions is defined by

$$DOSendth = DKLOS * p0S \tag{57}$$

This is a property of the endpoints, not of the entire history. For a cycle in which pnS=p0S, it vanishes even if work was performed and entropy was produced during the cycle.

Path distinguishability is defined on measures of histories:

$$AOSth = DKLOS, *PR \tag{58}$$

It answers the question of how distinguishable the observed forward history is from the observed time-reversed history. Under full observation and the conditions of § 7,

$$Aidth = th \tag{59}$$

For coarse-grained observation, the data-processing inequality gives

$$AOSthth \tag{60}$$

Coarse-graining can conceal part of the irreversibility, but it cannot increase the available KL distinguishability of the forward and reverse histories: this follows from the data-processing inequality because the same kernel OS, is applied to both measures. The external clock channel KCS, is also a coarse-graining and can therefore conceal temporal asymmetry together with indistinguishable events. Kawai, Parrondo, and Van den Broeck (2007) obtained a related phase-space bound on dissipation, while Roldán and Parrondo (2010) showed how irreversibility can be estimated from stationary observed time series. The equality between full path distinguishability and is an equality between two KL measures under the conditions of § 7, not an identity between the concepts of "information" and "entropy."

Nonequilibrium Free Energy and Distinguishability from Equilibrium

For a system in contact with a thermal environment,

$$Fp, = Up, -TbSp \tag{61}$$

For an isothermal transition between equilibrium states, the Jarzynski equality and Jensen's inequality imply [16]

$$\langle W \rangle thF \tag{62}$$

The mean dissipated work is defined as

$$Wdiss = \langle W \rangle th - F \tag{63}$$

For an isothermal system in contact with a single heat reservoir, with equilibrium initial and final distributions corresponding to the initial and final values of the protocol, and in the absence of additional entropy fluxes, this quantity is related to the mean dimensionless entropy production by

$$Wdiss = kBTbth \tag{64}$$

For a nonequilibrium distribution p relative to the equilibrium distribution corresponding to the same energy function E„ the nonequilibrium free energy can be written as

$$Fp = Feq + kBTbDKLp\| \tag{65}$$

Thus, KL distinguishability from equilibrium contributes to the available free energy of a particular physical system [14, 24]. This does not imply that arbitrary information is equivalent to work: the equality pertains to a specified distribution, energy function, and thermal environment.

Example: Bit Erasure

Consider a symmetric two-state memory. Before erasure,

$$p0 = 12, 12, \tag{66}$$

and after ideal erasure,

$$p1 = 1, 0 \tag{67}$$

The exact state (67) is an idealization; physically, it is attained as the limit of a protocol with a vanishingly small error probability. The entropy of the memory changes by

$$Smem = -kBln2 \tag{68}$$

In the reversible isothermal limit, the environment receives heat kBTbln2, while the memory receives

$$Q = -kBTbln2 \tag{69}$$

Therefore, the total entropy production is

$$= -kBln2kB - -kBTbln2kBTb = 0 \tag{70}$$

In the symmetric protocol considered here, reducing the memory entropy requires a minimum work input of kBTbln2, but in the reversible limit this reduction is fully compensated by the increase in the entropy of the environment. An irreversible implementation incurs additional dissipation and >0.

The example distinguishes three quantities:

1. the change in memory entropy;
2. the minimum work of erasure;
3. the entropy production of a particular protocol.

This example illustrates the standard symmetric form of Landauer's principle in the absence of accessible side information [21, 5], rather than asserting that erasing one bit always produces exactly kBln2 of total entropy.

Physical Implementation of Observation and External Clocks

The abstract channels OS and KCS, do not themselves perform work. A cost arises when a channel is implemented by a physical register A with its own states, memory, and dynamics. One then considers the joint space

$$XSXA \tag{71}$$

and the joint evolution ptx,a. Measurement is a correlation-generating transition

$$x, a0x, ay, \tag{72}$$

that implements the channel

$$OAa \mid x = PrAk + 1 = a \mid Xk = x,\ Ak = a0 \tag{73}$$

In the case of external clocks, the role of x is played by the state or label of the clock process C, while the transition aj−1aj constitutes a successful registration of the form (24). Without a physical change in the register, a signal may exist in the metasystem, but its tick does not enter the record accessible to the system. A channel without persistent memory may create a transient correlation, but it does not allow S to compare successive readings.

Once A is included within the physical boundary of the extended system, its own work, heat, entropy change, and entropy production can be defined. The cost of creating a correlation, the cost of signal transmission, the cost of reliably storing the result, and the cost of resetting the memory for reuse must be distinguished. Landauer's principle directly constrains logically irreversible memory erasure, rather than every act of discrimination or every clock tick as such.

Thus, thermodynamic cost pertains not to a channel as a mathematical mapping but to the chosen mechanism of its implementation. In the map (41), the observer and the clocks may remain outside the boundary of the thermodynamic system; an extended map for S+A+C must account for the balances of all included subsystems.

Object Types and Object Tokens

Let the recognition scheme be

$$R* = OS, , d, B, ; qtype, qtoken, h, \ell mintype, \ell mintoken \tag{74}$$

The components of the scheme are distributed between its two branches as follows:

Component	**Role**	**Branch**
OS	system observation channel	both
	maps an observed distribution or empirical measure to features	both
d	metric on the feature space Z	both
B={Ba}aA	catalog of type regions	both
	admissible deviation from a type region	both
qtype	common ensemble clock	type
qtoken	rule for selecting the clock qtoken of an individual history	token
h	size of the local averaging window	token
ℓmintype	minimum duration of a type manifestation	type
ℓmintoken	minimum token duration	token

Here Z is the feature space and A is the set of type indices; they are auxiliary structures rather than separate components of the tuple R*. The clock qtype must be common to the ensemble and independent of any particular realization; it may be the slice index k, the calibrated clock tC, or the mean Sint. The clock qtoken pertains to a specific history and may equal Sint, S|Cext, tC, or another monotone functional. If the catalog B is not specified, the construction identifies stable regimes, but assigning them to the same type requires a separate classification stage.

For an index segment I=i,j, its duration with respect to the selected clock is

$$\ell qI = qj - qi \tag{75}$$

Equation (75) is applied with q=qtype in the ensemble branch and with q=qtoken in the realization branch. Duration is therefore not a hidden absolute parameter of the scheme: the choice of clock is part of the conditions for recognizing an object.

Object Type

The ensemble feature representation is

$$zktype = OS * pkS \tag{76}$$

For a type a, define the set of clock-indexed slices on which the ensemble representation is recognized as belonging to that type:

$$Gatype = k : dzktype, Ba \tag{77}$$

An object type is specified by the region Ba together with the recognition scheme R*. Its ensemble manifestations are the maximal consecutive segments IGatype for which ℓqtypeIℓmintype. Thus, the stability of a distribution defines a reproducible macroregime rather than an individual thing.

Object Token

For an individual realization =x0,e1,x1, and an observed history ykOS|xk, set qk=qtokenk and introduce the causal window

$$Jk, hq = jk : 0qk - qjh \tag{78}$$

If the window is nonempty, the local empirical measure of observations is

$$k, h\ , q = 1Jk, hqjJk, hqyj \tag{79}$$

The parameter h sets the scale of the selected clock over which rapid fluctuations are suppressed. Under irregular sampling, the equal weights in (79) may be replaced by a prespecified sampling measure. The corresponding feature representation is

$$zk, htoken, q = k, h\ , q \tag{80}$$

For a type a, set

$$Ga, token, h; q = k : dzk, htoken, q, Ba \tag{81}$$

An object token of type a is a maximal consecutive segment I of this set for which ℓqIℓmintoken. If the regions Ba are not specified, it is appropriate to speak only of a stable realization-level regime, not of a token of a particular type.

Thus, an object type and an object token are distinct forms of process stability. Ensemble stability does not guarantee the stability of every realization, while a stable token does not require the entire ensemble to be stationary. Under a deterministic description or a degenerate distribution, these definitions may coincide, but in general they differ.

The stability of an object does not imply that its internal time has stopped. As long as zk,htoken,q remains within the type region Ba, the realization may undergo many of its own transitions and accumulate a substantial Sint. A thing persists not because nothing within it changes, but because its changes do not destroy its recognizable organization. The converse is also possible: a token may continue to exist with respect to external time even though its internal event counter temporarily ceases to increase.

The same object segment therefore has at least two substantively different durations:

$$\ell intI = Sintj - Sinti, \qquad \ell extI = S \mid Cextj - S \mid Cexti \tag{82}$$

The first shows how many of its own changes the token has accumulated; the second shows how many accessible external ticks have elapsed in the metasystem. Their mismatch is part of the description of the object, not a defect of the clocks.

Gaps, Division, and Robustness of the Criterion

The strict condition (81) terminates a token when it leaves the type region. If identity is to be preserved across a brief gap or an absence of observations, an additional continuation predicate CRIi,Ij is required to link two stable segments. Such a predicate does not follow from the metric d alone: it expresses a separate rule of identity and may account for the admissible duration of the gap according to the selected clock, causal reachability, or conserved quantities.

Division and merger likewise cannot be described by a single interval. They form a directed graph of token provenance: a vertex with several outgoing edges represents division, whereas a vertex with several incoming edges represents merger. A linear worldline is a special case of this structure.

Objecthood should not depend on a single precisely chosen threshold. Let I={I1,,IN} be the segmentation of a history at threshold . Number its segments in the order of the original history,

$$maxIi < minIi + 1, \qquad i = 1, , N - 1$$

Only segments of the same chronological rank may be matched; a later segment in one segmentation cannot be matched with an earlier segment in another. Define the distance

$$dsegI, I' = \{1iNdHq * Ii, Ii', N = N' = N, +, NN' \tag{83}$$

Recognition is robust over a range of thresholds E if

$$sup,' EdsegI, I' \tag{84}$$

Here q* denotes qtype or q in the corresponding branch, and

dHqI,J=dH{qminI,qmaxI},{qminJ,qmaxJ}

is the Hausdorff distance between the segment boundaries mapped to the clock coordinate. If q is not injective on the set of all admissible boundaries, dHq is a pseudometric. Because it takes the value + when the numbers of segments differ, dseg is, in general, an extended pseudometric. Quotienting by the zero-distance relation and requiring q to be injective on all admissible boundaries yields an extended metric; it is finite on each sector with a fixed number of segments. The condition controls not only the proximity of the boundaries but also the number of identified tokens, thereby distinguishing robust objects from artifacts of fine-tuning the threshold.

The dependence of an object on R* does not entail arbitrariness. A recognition scheme is constrained by measurement reproducibility, robustness under small perturbations, and the ability of the identified patterns to support testable predictions. In this sense, the definition is close to Dennett's notion of "real patterns," but it is furnished with an explicit trajectory-level criterion [9].

Positioning the Proposed Framework

The process philosophies of Whitehead and Rescher assert the priority of becoming and process over substance [31, 25]. The present work adopts this general shift but articulates it through a stochastic specification of transitions, trajectory functionals, and an operational recognition scheme.

In distributed-systems theory, the happened-before relation distinguishes the causal partial order of events from logical-clock readings [20]. Event structures formalize causality, conflict, and concurrency without requiring a global time [32]. The proposed construction uses this distinction as an ontological layer, while adding internal measures of change, a physical external-clock channel, and thermodynamic functionals of calibrated trajectories.

At a more applied computational level, Active Transaction Graphs [33] provide an executable semantics for mediated transition histories, distinguishing state results, execution traces, and ledger effects. The present framework operates at a more primitive physical level: it begins with a causal partial order and introduces thermodynamic functionals only when an energy model, a calibrated clock, and the corresponding physical prerequisites have been specified.

The causal-set program treats a locally finite causal order as a candidate for the fundamental structure of spacetime [6]. No such physical hypothesis is adopted here: EM is a labeled causal order of the events in the model, while discreteness pertains to operationally distinguishable and recorded events. Any connection to quantum gravity would require a separate formalism.

Relational approaches to time in physics hold that temporal structure is defined by correlations with a selected physical clock rather than by an external parameter: the Page–Wootters mechanism recovers dynamics from correlations between a subsystem and a clock subsystem within a stationary whole [23], relational quantum mechanics relativizes states to the observing system [27], and Machian analyses eliminate absolute time altogether [4]. The central relation (85) is a classical, operational counterpart of this family: external time is likewise borrowed from a designated clock process, but the mechanism here is an explicitly modeled, generally lossy delivery channel and register, and the construction neither requires nor implies a timeless global state or any quantum formalism.

Constructor theory describes physics in terms of possible and impossible transformations [10, 11]. The affinity here lies in the priority given to transformation; the difference lies in the focus on a particular realized trajectory, its cost, irreversibility, and observable distinguishability.

Categorical theories of processes provide an abstract language for composing processes and morphisms [1, 7]. The proposed framework is less general algebraically, but it connects the process description directly to probabilistic and thermodynamic quantities.

Ontic structural realism shifts the emphasis from individual objects to relations and structure [19]. Here the object receives a more specific definition: it is a stable feature regime of a trajectory relative to R*.

Work on semantic information investigates the relation between information, a system's continued existence, and its goals [18]. In the present article, information is used in the narrower statistical sense of distinguishability between distributions and trajectory measures. Semantic value, agency, and purposiveness do not follow from this definition.

A complementary, goal-relative notion of noise is developed in [34], where a capacity-efficient good regulator is shown to factor through a projection that suppresses distinctions irrelevant to admissible action. This notion should be distinguished from the channel-relative concept used in the present article: here, indistinguishability is defined statistically with respect to an observation kernel and a specified pair of probability measures, without presupposing a goal or an action set.

Stochastic thermodynamics under coarse graining shows that hidden microstates can alter observed balances and induce effective memory [13]. The present work does not identify this result with a theory of clocks, but it does support the requirement that the channel and the boundary of the observed system be specified explicitly.

A domain-specific control-theoretic realization of this boundary dependence is developed in [35], where the boundary of an organizational digital twin is represented through actor membership, interface permeability, and operational dynamics. Although that application concerns firms rather than physical clock records, it illustrates the general point that boundary selection changes the modeled state variables, available observations, admissible controls, and conditions of viability.

Scope and Limitations

First, the relation $\prec M$ is taken as a primitive of the framework rather than derived from thermodynamics or changes of state. This removes circularity from the definition of time, but makes causal precedence an explicit postulate of the model.

Second, local finiteness and discrete clock events establish the discreteness of operationally accessible readings. They do not imply that physical spacetime is fundamentally discrete. A continuous coordinate t requires an additional measure, calibration, and interpolation, as well as an explicit notion of convergence if a continuum limit is claimed.

Third, a partial order does not determine a unique global clock. Selecting the chain C excludes concurrent events, while any scalar linear extension discards part of the structure of causal independence. Event conflict is not modeled separately in the present causal order. External time is therefore always relative to a chosen reference process, boundary, and channel.

Fourth, KCS, may omit, merge, delay, reorder, and distort ticks. Under noise, a reading should be represented by a conditional distribution or an uncertainty interval. Back-action of S on C, asymmetric delays, and synchronization among different systems require a separate model.

Fifth, the internal event measure is defined directly for locally finite discrete histories. It depends on the system boundary and the criterion for distinguishing states, is not work, information-theoretic distinguishability, state entropy, or entropy production, and does not preserve intervals between events. Diffusive and smooth trajectories require a metric, path variation, or another clock functional.

Sixth, the general map does not make thermodynamic components mandatory. Work, heat, and physical entropy production are defined only when an energy function, a thermal environment, and conditions linking the dynamics to the energetics are available. They cannot automatically be transferred to a text, an institution, or a computational model.

Seventh, the independence of thermodynamic functionals from a linear execution rests on Proposition I.1 and Corollary I.1: the protocol–system comparability axiom (10), the passage to the energy skeleton, and the specification of the calibrated clock record independently of linearization. If causally incomparable events can simultaneously modify x or , or if the temporal assignment itself depends on an arbitrary linear extension, the causal model must be enlarged or the relevant layer must be defined relative to an explicitly chosen execution.

Eighth, the equality between entropy production and the KL divergence of forward and reverse trajectories is used only under the conditions of § 7. The conditional equality Aid= is not an identity of information and entropy: coarse-graining the channel or changing the pair of compared measures generally destroys it. Partial observation may render the dynamics non-Markovian; in that case, the state space must be enlarged with hidden variables, or the trajectory measures and reverse process must be specified separately.

Ninth, common dependence on a transition history does not license the identification of clock readings, work, information-theoretic distinguishability, and entropy production. Relations among them are calibrations, inequalities, or model-dependent trade-offs requiring additional physical parameters; there is no universal conversion rate.

Tenth, the scheme R* does not provide a unique "true" partition of the world into things. It makes the channel, clock, scale, and recognition criterion explicit and testable. Different schemes may identify different but simultaneously stable objects.

Finally, a channel is not a subject of experience, and the article does not prove the metaphysical primacy of transition as a theorem of physics. Describing an agent would require memory, an action-selection rule, feedback, and a criterion for selecting protocols; consciousness lies beyond the scope of this work.

Conclusion

A transition is not reducible to the replacement of one state by another and does not presuppose an underlying continuous interval. The framework is grounded in a locally finite causal partial order of events. Along a local history, a system's internal time accumulates as its own states change. External time is likewise event-based: the selected reference process C changes within the metasystem M, while the system S receives only a physically transmitted and partially observed record of those changes.

The central relation is

$$PS \mid Crec = KCS, *PCclock, S \mid Cext = Ftick, \qquad CM, \quad CS = \emptyset \tag{85}$$

In other words, the external-clock reading available to the system is a functional of the transmitted record of a selected clock process, not a reading of an absolute metasystem time. The incompleteness of the channel makes this coordinate local and relative; registering a tick requires an internal change in the memory of S. A real-valued calibration turns the count into a duration, while a continuous coordinate t is introduced as an effective interpolation or obtained in a separately specified scaling limit.

The common transition substrate does not make its valuations identical. Internal event time is a clock functional determined by a boundary, resolution, and weighting rule. Work is an energetic functional determined by an energy model, protocol, and work channel. Information-theoretic distinguishability is a relation between specified measures relative to an observation channel. Entropy production is a thermodynamic interpretation of forward–reverse path asymmetry under the conditions of § 7. A stationary equilibrium system with nonzero dynamical activity at fixed protocol and under detailed balance may accumulate internal ticks while work, directional distinguishability, and entropy production all vanish. Conversely, a reversible quasistatic protocol may perform nonzero work while entropy production tends to zero. These separations — the separation principle of § 5.1 — rule out any universal identity among time, work, information-theoretic distinguishability, and entropy production.

Their relations instead take the form of conditional balances and bounds. Landauer's principle concerns the cost of specified logically irreversible operations, not every act of observation or every tick. Thermodynamic speed limits relate achievable change, calibrated duration, dynamical activity, and dissipation without identifying them. Coarse-graining bounds observable path distinguishability by full path distinguishability. The physical model—not the concept of transition alone—determines which such relations exist and what scales enter them.

The principal ontological proposal of the article concerns the thing. At the ensemble level, an object type is defined by the stability of a macroregime; at the level of an individual history, an object token is defined by the stability of a realization relative to an explicitly selected clock and scale. In both cases, objecthood consists not in the absence of change but in preserving the recognizable organization of a process. The same token thereby carries two irreducible durations — the internal count of its own changes and the accessible external count (82); their mismatch is part of the description of the object, not a defect of the clocks:

Transition is taken as primitive; a thing is defined as its stable organization.

The resulting picture is plural without being fragmented: causal order provides the primitive structure, whereas clocks, duration, work, information-theoretic distinguishability, and entropy production are distinct typed characteristics defined relative to a common process specification. Their irreducibility makes physical comparison meaningful, while the common specification makes their joint description coherent.

Appendix A. Delivery, Registration, and Decoding of External Ticks

Attempt Record and Memory Update

A complete protocol-level implementation of the clock channel refines the output record =1,2„ the observed signals yj, and the register states rj:

$$KCS, \mid C, \qquad ck \prec Mj \quad kAj, \qquad rj = Umemrj-1, yj, \qquad wj = Dwrj-1, yj, rj \tag{A1}$$

Here Aj is the set of source ticks represented by the jth attempt. For a genuine nonaggregated signal, Aj={j}, where j is the index of the tick that generated it; for an aggregated message, Aj>1; for a false signal, Aj=∅ and j is undefined. The decoder is chosen such that wj0, and, upon successful registration rjrj−1, a tick included in the count has wj>0. For an unsuccessful or rejected attempt, one sets wj=0.

An omission corresponds to a tick that belongs to no Aj; duplication is the occurrence of one index in several sets; aggregation is an Aj containing more than one element. These cases are properties of a particular channel, not of the underlying causal order.

Incomplete Knowledge and Message Order

For a noisy or incomplete channel, the system's state of knowledge may be a distribution over the hidden source reading. Set Jn=max⋃jnAj, with max∅=0, and let c0 denote the initial clock state before the first represented tick, with Cintc0=0. The complete register history available by step n generates the σ-algebra FnR=r0,y1,r1„yn,rn. Then

$$S \mid C, nB = PrCintcJnB \mid FnR \tag{A2}$$

The raw record preserves source order if j<j′ implies k<k′ for every kAj and k′Aj′. If this condition is violated, the sequence must be reconstructed by a separate protocol, for example using stored tick numbers or causal labels. If rn is a sufficient state of the channel's entire memory, conditioning on FnR in (A2) may be replaced by conditioning on rn. The filtering distribution does not assume that the external states of the metasystem are fully observable to S.

Decoding Physical Duration

The calibration of the output register is specified by a separate duration decoder. Let jerr0 denote its output in a registered case that is not an exact decoding of a genuine single or aggregated signal.

$$j := Drj-1, yj, rj0,$$
$$j = \begin{cases} j, & rjrj-1,\ Aj = \{j\},\ \text{signal accepted and decoded exactly,} \\ kAjk, & rjrj-1,\ Aj > 1,\ \text{aggregate accepted and decoded exactly,} \\ jerr, & rjrj-1,\ \text{signal accepted, all other cases,} \\ 0, & \text{otherwise.} \end{cases} \tag{A3}$$

In an ideal channel, Aj={j} and j=j>0. For a registered false or distorted signal, the value jerr is specified by the error model; such a contribution is a reading error rather than the duration of a genuine source tick. Thus, the positivity of the original increments k is compatible with zero decoded increments for unsuccessful attempts.

Appendix B. Worked Example: A Two-State System with a Lossy External Clock

Setup

Let the system have two states, a and b, observed through the identity channel, with unit weights in (17). The clock process emits five ticks with unit source calibration in (26), so its source coordinate advances by one unit per tick. The delivery channel of (A1) delivers the first and third ticks exactly, loses the second tick (it belongs to no provenance set), and aggregates the fourth and fifth ticks into a single message that is decoded exactly: the weight decoder of (A1) returns a decoded weight of 2 (one unit per represented tick) for use in (20), and the duration decoder of (A3) returns a decoded increment of 2 for use in (27). All three delivery attempts result in successful registrations, so the register changes three times. The system performs five intrinsic transitions (a→b→a→b→a→b) and then remains in b. The interleaving of intrinsic transitions, source ticks, and registrations along the selected execution is shown in Table B1; every entry follows by direct application of the indicated equations. Explicitly, t*=0, k=1 and wC=1 for every source tick, and the provenance sets are A1={1}, A2={3}, A3={4,5}.

Table B1. Event log of the worked example. Columns give, after each step: the system state; the internal event time (17); the number of successful registrations and the source tick count entering the undercount bound (23); the external time (20); the source coordinate (26); and the decoded coordinate (27). Each row is an observation cut along the selected execution and may group elementary events.

Step	**Event**	**x**	**Sint (17)**	**NS|Cext / NCint (23)**	**S|Cext (20)**	**tC (26)**	**tS|C (27)**
0	initial state	a	0	0 / 0	0	0	0
1	tick 1 delivered; registration 1	a	0	1 / 1	1	1	1
2	intrinsic transition a→b	b	1	1 / 1	1	1	1
3	intrinsic transition b→a	a	2	1 / 1	1	1	1
4	tick 2 emitted and lost	a	2	1 / 2	1	2	1
5	intrinsic transition a→b	b	3	1 / 2	1	2	1
6	intrinsic transition b→a	a	4	1 / 2	1	2	1
7	intrinsic transition a→b	b	5	1 / 2	1	2	1
8	tick 3 delivered; registration 2	b	5	2 / 3	2	3	2
9	ticks 4 and 5 emitted; aggregated in channel	b	5	2 / 5	2	5	2
10	aggregate delivered; registration 3 (w3=2, 3=2)	b	5	3 / 5	4	5	4

Clocks and the Channel

The log exhibits the clock-theoretic claims of § 3 on concrete numbers. The undercount bound (23) holds with room to spare: three registrations against five source ticks. The decoded coordinate (27) terminates at 4 while the source coordinate (26) terminates at 5; the missing unit is exactly the lost second tick, which cannot be recovered from the delivered record and the stipulated decoder alone — the record is partial in precisely the sense of § 3.2. The aggregated message illustrates the difference between counting registrations and accumulating decoded weight: the third registration is a single register change but carries decoded weight 2 for (20) and decoded duration 2 for (27), so both the external time and the decoded coordinate jump from 2 to 4 in one step. An exact instance of (25) occurs on steps 8–10, where the internal increment is 0 while the external increment is 2. Its converse occurs on steps 1–7, where the corresponding increments are 5 and 0.

Token Segmentation and Robustness

Take as observations the eleven row states of Table B1, including the initial row, indexed by the row number k = 0, ..., 10, with observed sequence a, a, b, a, a, b, a, b, b, b, b, and let the token clock in (74) be the step index with averaging window h = 2, so that the causal window (78) contains at most the last three observations. Take the feature map Φ of (80) to be the empirical frequency of the state b in the window, and let d be the Euclidean metric on the feature line, with the induced point-to-set distance d(z, B) = inf over b ∈ B of |z − b| used for the recognition regions: the alternating regime [1/3, 2/3] and the frozen regime [0.9, 1]. The resulting feature values are 0, 0, 1/3, 1/3, 1/3, 1/3, 1/3, 2/3, 2/3, 1, 1.

At tolerance $\varepsilon = 1/12$, with a common minimal token duration of one step for both regimes, the recognized sets (81) are {2, ..., 8} for the alternating regime and {9, 10} for the frozen regime, yielding two object tokens: an alternating token on the segment [2, 8] and a frozen token on [9, 10]. The segmentation is robust in the sense of (84): the smallest feature-to-region distance that could change any membership is 0.9 − 2/3 ≈ 0.233, so over the entire threshold range E = [0, 0.2] the segmentation is unchanged and the segmentation distance (83) is zero — the two tokens are not artifacts of a fine-tuned threshold.

The dual durations (82) complete the picture and are read directly from the two θ columns of Table B1. The alternating token accumulates four intrinsic transitions while the accessible external time advances by one unit: its internal duration is 5 − 1 = 4 against an external duration of 2 − 1 = 1. The frozen token accumulates no intrinsic transitions while the external register advances by two units: internal duration 0 against external duration 4 − 2 = 2. At the level of recognized objects, the frozen token realizes (25) exactly, whereas the alternating token exhibits the opposite clock imbalance, ℓint>ℓext, without zero external duration: a thing that persists by the world's clock while its own clock stands still, and a thing that lives fast by its own clock while the delivered clock barely moves. The mismatch is part of the description of each token, not a defect of either clock.

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ONTOLOGY OF TRANSITION—PART II

The Unidentifiable Clock: Reconstruction Limits and Gauge Freedom of External Time under Lossy Delivery

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Abstract

The three-part scientific work *Ontology of Transition* develops the central thesis that transition, rather than the static thing, is the primary unit of description, and that a system's operational time is constructed from physical records of transitions communicated through finite, imperfect channels. This article is Part II of that work.

Part I [1] argued that a system's accessible clock coordinate is a functional of a physically transmitted, generally lossy record of a reference process, not a reading of the source itself — a conceptual organization of established results. Part II turns that qualitative thesis into quantitative theorems about the same model, using Part I's record, register, and decoder without modification. First, for an erasure channel with known tick count, the minimax mean-squared error of reconstructing the source coordinate from the delivered record is exactly n p σ^2 — the number of source ticks times the loss probability times the variance of the calibration increments — achieved by the conditional-mean decoder; in particular, the source coordinate is perfectly recoverable through an arbitrarily lossy channel iff calibration is deterministic. Second, when the tick count itself is unknown, an additional floor of μ^2 n p/(32 π q) is proved for deterministic calibration by a self-contained two-point argument. Third, and centrally: for the aggregating channel, the pair (loss rate, calibration law) is never identifiable from the record law alone for any positive loss rate — it is determined only up to a one-parameter semigroup orbit of geometric compounding, which contains at most one geometrically indecomposable representative; when the physical calibration law is indecomposable, it is that representative. Operationally: an observer behind the channel cannot distinguish a fast source behind a lossy channel from a slow source behind a clean one. The rate of external time is gauge, and the gauge freedom is generated by geometric compounding. Fourth, the reader itself is bounded: for a register in a nonequilibrium steady state, the relative variance of the decoded coordinate is at least 2/reg by the thermodynamic uncertainty relation — the accessible clock thus carries two distinct irreducible error floors, expressed in different metrics and attributed to different physical mechanisms — one information-theoretic (paid in lost calibration fluctuation), one thermodynamic (paid in register dissipation). Fifth, objecthood itself has a critical point: for the canonical alternating source read through the aggregating channel, robust token recognition in the sense of Part I's recognition scheme survives exactly up to a critical loss rate pc = $(\delta+\varepsilon)/(1-\delta-\varepsilon)$, set by the channel's parity bias, above which the faster type dissolves while the frozen type persists at any loss. Provenance side-information collapses the gauge orbit and restores identifiability, exactly as the parent framework's message-order protocol suggests. All results are verified numerically.

Keywords: *external clocks; lossy delivery; reconstruction limits; gauge freedom; thermodynamic uncertainty; object recognition*

1. Setting

We work inside the model of Part I [1, §3.2, App. A], keeping its notation; the record-as-message viewpoint descends from the event-ordering tradition of [2]. A related formal treatment of mediated interaction, execution traces, and observational metadata is developed in [13]. The setting: a clock process C emits ticks c_1, c_2, ... with calibrated increments k > 0 (eq. (26) of [1]); a delivery channel produces a record of attempts with provenance sets Aj (eq. (A1)); the system's decoded coordinate accumulates decoded increments (eq. (27)). Throughout, the calibration increments k are i.i.d. from a law F on $(0, \infty)$ with mean $\mu < \infty$ and variance $\sigma^2 < \infty$. Two channel archetypes from [1, App. A] are formalized:

(E) Erasure channel Each tick is independently delivered with probability q = 1 − p and lost with probability p (its index belongs to no Aj). Delivered ticks are decoded exactly and in source order. The record after n source ticks is the ordered list of delivered increments; the observer sees the list and its length N, nothing else.

(A) Aggregating channel Ticks are processed in order; each is independently buffered with probability p or flushed with probability q, a flush delivering one message that carries the sum of all buffered increments since the previous delivery, plus its own, decoded exactly (eq. (A3), aggregate case). Message values are then i.i.d. compound-geometric: the m-th block has size distributed Geom(q) on $\{1, 2, \dots\}$, and value distributed G = m1 q pm−1 F*m.

The target of reconstruction is the source coordinate tC(n) = kn k (eq. (26) of [1]). The observer's information is the σ-algebra generated by the record — the register history FnR of [1, eq. (A2)].

2. The exact reconstruction floor

Theorem II.A (price of a lost tick). Under channel (E) with known n and known (p, F), for every estimator t measurable with respect to the record,

$$Et - tCn2np2$$

with equality for the conditional-mean decoder t* = SN + (n − N) μ, where SN is the sum of delivered increments and N their number.

Proof. Write tC = SN + M, where M is the sum of the lost increments. The delivery mask is independent of the increment values, and the increments are i.i.d.; hence, conditionally on the record (which determines N and the delivered values), M is a sum of n − N i.i.d. draws from F, independent of the delivered values. Therefore E[M | record] = (n − N) μ and Var(tC | record) = Var(M | record) = (n − N) σ^2 almost surely. For any record-measurable t, E[(t − tC)^2] $\geq$ E[Var(tC | record)] = σ^2 E[n − N] = n p σ^2, and the conditional mean attains it. □

Corollary II.A1 (dichotomy of recoverability). tC is exactly recoverable from the record for some (equivalently, every) p $\in$ (0, 1) iff $\sigma^2 = 0$, i.e. iff calibration is deterministic. Losing ticks is not the same as losing time: with a degenerate calibration the decoder t reconstructs tC without error at any loss rate, because each lost tick carries a known duration. What a lossy channel irrecoverably destroys is exactly the fluctuation of calibration, at a rate of σ^2 per lost tick.

Remark. Var(tC | record) is precisely the variance of the filtering distribution Π of [1, eq. (A2)]; Theorem II.A computes it in closed form for channel (E) and shows the parent framework's "state of knowledge" is not a metaphor but the exact Bayes object.

Numerical check Appendix N.1: empirical MSE matches the floor to 0.3%; the degenerate calibration gives identically zero error.

3. Unknown tick count

Lemma II.B1 (local bound). maxi P(Bin(n, q) = i) $\leq$ $\sqrt{}$(π / (8 n q p)), where p = 1 − q.

Proof. By Fourier inversion, every point mass is bounded by (1/2π)∫−π^π |ϕ(t)| dt, where ϕ is the characteristic function. For the binomial, |ϕ(t)|^2 = 1 − 2qp(1 − cos t) $\leq$ exp(−2qp(1 − cos t)), hence |ϕ(t)| $\leq$ exp(−n q p (1 − cos t)). Using 1 − cos t $\geq$ 2t^2/π^2 on [−π, π] and extending the Gaussian integral to the line, (1/2π)∫ exp(−2nqp t^2/π^2) dt = (1/2π) · 3/2/$\sqrt{}$(2nqp) = $\sqrt{}$(π/(8nqp)). □

Theorem II.B (counting floor; local minimax). For channel (E), suppose $\sigma^2 = 0$, the tick count n is unknown to the observer, and n p $\geq$ 2πq. Then every record-measurable estimator obeys the bound below. The explicit constant 1/(32π) is not claimed to be optimal.

$$supn'n, n + kEt - tCn'22np/32q$$

Proof. With $\sigma^2 = 0$ the record law depends on n only through N ~ Bin(n, q) (delivered values are i.i.d. F under both hypotheses and carry no information about n). Step 1 (shift TV): for the unimodal binomial, TV(X, X+1) = $\frac{1}{2}$ Σ_i $|p_i - p_{i-1}|$ = maxi p_i by telescoping over the mode, so TV(X, X+j) ≤ j · maxi p_i; by the coupling Bin(n+k, q) =d Bin(n, q) + Bin(k, q) and convexity, TV(Bin(n,q), Bin(n+k,q)) ≤ E[Bin(k,q)] · maxi p_i ≤ kq · $\sqrt{}(\pi/(8nqp))$ = k · $\sqrt{}(\pi q/(8np))$ by Lemma II.B1. Step 2 (two points): choose k = $\lfloor\sqrt{}(2np/(\pi q))\rfloor$; the hypothesis np ≥ 2πq gives $\sqrt{}(2np/(\pi q)) \geq 2$, hence k ≥ $\frac{1}{2}\sqrt{}(2np/(\pi q))$ and TV ≤ $\frac{1}{2}$. The targets differ by kμ; Le Cam's two-point bound gives max risk ≥ $(k\mu/2)^2 \cdot (1 - TV)/2 \geq k^2\mu^2/16 \geq \mu^2 np/(32\pi q)$. □

Remark (combining the floors). For unknown n, Theorem II.A's floor persists at every σ^2: an estimator ignorant of n is admissible in each known-n problem, so the supremum over n of the risk is at least n p σ^2 pointwise in n. By the Bayes decomposition $E(t - tC)^2 = E(t - E[tC \mid record])^2$ + E[Var(tC | record)], the calibration floor lives in the second term for every σ^2, while the counting floor of Theorem II.B bounds the first term as proved here only at $\sigma^2 = 0$ (the two-point argument uses the determinism of the target; for $\sigma^2 > 0$ the target's own fluctuation $\sigma\sqrt{}$n is of the same order as the separation). Hence risk ≥ n p σ^2 always, and additionally risk ≥ μ^2 n p / (32πq) at $\sigma^2 = 0$; a sharp counting floor for $\sigma^2 > 0$, and whether the floors add, are left open.

Numerical check Appendix N.2: Lemma II.B1 and the shift bound verified in four regimes; the prescribed k* yields TV = 0.30 ≤ $\frac{1}{2}$.

Combined with Theorem II.A: the proven floors for unknown n are n p σ^2 at every σ^2, and additionally c μ^2 n p/q when calibration is deterministic; their joint sharp form is open. In both regimes the channel destroys information at a rate linear in n p.

4. The compounding gauge: non-identifiability of the clock channel

The deeper question is not reconstruction of a trajectory but identifiability of the mechanism: does the record law determine (p, F)?

Lemma II.C0 (composition of geometric blocking). For block-size generating functions ga(z) = a z / (1 − (1−a) z) on {1, 2, . . . }: gq″ ∘ g = gq″ . Consequently, a Geom(q″)-sum of independent Geom(θ)-blocks is a Geom(q″ θ)-block.

Proof. Direct computation: gq″(g(z)) = q″ θ z / (1 − (1 − q″ θ) z). □

Theorem II.C (gauge orbit of external time). Under channel (A), let G(p, F) denote the law of message values. Then for every q″ ≥ q (i.e. every less lossy channel p″ ≤ p),

$$Gp, F = Gp'', F'', \qquad F'' = Geomq/q''F$$

Hence the record law determines (p, F) only up to the totally ordered orbit

$$Op, F = p'', Geomq/q''F : 0p''p$$

whose endpoint p″ = 0 declares the channel lossless and attributes all blocking to the source. Conversely, two representations (p_1, F_1), (p_2, F_2) of the same record law with $q_1 < q_2$ satisfy F_2 = Geom(q_1/q_2)∘F_1. Therefore:

(i) (p, F) is never identifiable from the record law when p > 0;

(ii) the orbit contains at most one representative whose calibration law is geometrically indecomposable (not itself a nontrivial geometric compound), and if F is geometrically indecomposable then (p, F) is that unique maximal-loss representative. The orbit O(p, F) exhausts the equivalence class of the record law iff F is geometrically indecomposable; for decomposable F the class extends to representations with $p'' > p$.

Proof. In Laplace transforms, G = q F / (1 − p F). For $q'' \geq q$ set $F'' = \theta$ F / (1 − (1−θ) F) with $\theta = q/q'' \in (0, 1]$; by Lemma II.C0 this is the transform of the Geom(θ)-compound of F, a bona fide law on (0, ∞), and substitution gives q'' F'' / (1 − p'' F'') = G. For the converse, solving $q_1F_1/(1-p_1F_1) = q_2F_2/(1-p_2F_2)$ for F_2 yields $F_2 = \theta$ F_1 / (1 − (1−θ) F_1) with $\theta = q_1/q_2$, i.e. F_2 = Geom(q_1/q_2)$\circ F_1$, which is a nontrivial compound whenever $q_1 < q_2$. Claim (ii) follows: if F_2 were also geometrically indecomposable, the displayed identity forces $q_1 = q_2$ and $F_1 = F_2$. □

Corollary II.C1 (rate of external time is channel-relative). Since the mean message value is μ/q, an observer behind channel (A) cannot distinguish "fast source, lossy channel" from "slow source, clean channel": every point of the orbit is an observationally complete description. In the vocabulary of [1], the accessible external clock fixes duration only relative to a chosen decomposition into source and channel; the decomposition itself is gauge. This sharpens the parent paper's claim that external time is relative to the triple (S, C, K): the (C, K) split inside the triple is not empirical but conventional, up to the geometric-compounding semigroup action.

Terminology Here gauge means observational redundancy among parameterizations of the same record law. It is an analogy to gauge freedom as redundancy of description, not a claim of a local symmetry of the underlying dynamics.

Corollary II.C2 (what breaks the gauge). Any side-channel that reveals provenance — stored tick numbers or causal labels in the sense of [1, App. A.2] — collapses the orbit to a point: with singleton provenance sets observable, N and the delivered indices are known and (p, F) is identifiable (F from delivered values, p from index gaps). Identifiability of the clock is thus purchased exactly by the metadata whose transmission the parent framework prices separately (§11 of [1]).

Numerical check Appendix N.3: two orbit points are statistically indistinguishable (KS p = 0.50); the transform identities hold to machine precision.

5. The thermodynamic floor of the reader

The thermodynamic cost of clock precision is studied theoretically and experimentally in [5, 6, 10, 11]. The theorem below transfers the current-precision bound to the reader's side of the channel, where the parent framework's separation of source, channel, and register makes the attribution exact.

Theorem II.D (register-precision bound). Let the register be a finite Markov jump process in a nonequilibrium steady state satisfying local detailed balance, whose net registration count N(t)—forward minus reverse transitions across designated registration edges—is a time-antisymmetric integrated empirical current with mean $\langle N \rangle > 0$, and let the register's total entropy production over the observation window be reg (in units of kB). Assume the decoded increments wj are i.i.d. with mean w > 0, independent of the registration process. Then the decoded coordinate t = jN wj satisfies

$$Vart/\langle t\rangle 2VarN/\langle N\rangle 22/reg$$

Proof. By the law of total variance with w independent of N: Var(t) = ⟨N⟩ Var(w) + w^2 Var(N) $\geq$ w^2 Var(N), while ⟨t⟩ = w ⟨N⟩; dividing gives the first inequality. The second is the finite-time thermodynamic uncertainty relation for integrated currents in a NESS under local detailed balance [4, 7]. □

Corollary II.D1 (two irreducible floors). The accessible clock coordinate carries two distinct irreducible error budgets, expressed in different metrics and paid to different parties. The channel floor of Theorem II.A is information-theoretic: n p σ^2 of variance destroyed per window, payable in lost calibration fluctuation, independent of any thermodynamics. The register floor of Theorem II.D is thermodynamic: relative variance at least 2/reg, payable in the reader's own dissipation, independent of source quality and channel fidelity. The two budgets constrain different observables through different mechanisms; they become additive only within an explicitly specified joint error model, and neither can be traded against the other. In the vocabulary of [1]: partial observability and the physical cost of observation (§11) bound the same coordinate from two irreducible directions.

Scope of the current observable A one-sided count of registration events is activity-like and is not covered by the thermodynamic uncertainty relation used here; such an observable requires an appropriate kinetic uncertainty relation. The literal success count may be identified with N only in an effectively unidirectional registration model or when reverse events are negligible on the stated scale.

Remark (scope). The independence hypothesis is substantive. It holds for erasure-type registers — channel (E) with exact decoding is the model case, since there each decoded weight is a fresh draw from F regardless of how many registrations the window contains. For the aggregating channel (A) it fails by construction: wj is a block sum, and over a fixed window larger blocks mean fewer registrations, so decoded weights and the counting current are negatively correlated. Theorem II.D therefore does not apply to channel (A) as stated; the aggregating case is open, and any extension must replace the total-variance step by one that tracks the weight–count covariance.

Remark (attribution). The bound involves reg only. A perfect source read through a perfect channel by a dissipationless register in steady state would still be TUR-bound; conversely, an arbitrarily dissipative register cannot repair the channel floor of Theorem II.A. The parent framework's insistence on separating source, channel, and register (eqs. (19)–(24), (71)–(73) of [1]) is exactly what makes this attribution well-posed.

Numerical check Appendix N.4: a unicyclic register realizes the full chain of inequalities within 4% of TUR saturation.

6. The dissolution of objects

We now give the recognition scheme R* of [1, §12] its first sharp threshold. Take the canonical alternating source: the emitted symbol flips at every source step (the alternating regime of [1, App. B] in its pure form). (Throughout this section q = 1 − p is the flush probability of channel (A).) Read it through channel (A), observing the current symbol at each delivery. Formally, each delivered message is an observation cut in the sense of [1, Table B1], and the observed symbol is the decoded register reading of [1, App. A] immediately after the aggregated registration — for a symbolic source, the source symbol at the block's final tick.

$$f := 1/(1 + p)$$

Lemma II.E1 (parity lemma). Successive delivered observations flip iff the intervening block size is odd; the flip indicators are i.i.d. Bernoulli(1/(1+p)).

Proof. Block sizes are i.i.d. Geom(q) on $\{1, 2, \dots\}$, and an alternating source changes symbol over a block iff the block is odd. P(m odd) = j0 q p2j = q/(1 − p^2) = 1/(1+p); independence is inherited from the blocks. □

As p → 0 the observed sequence alternates perfectly; as p → 1 the flip probability tends to $\frac{1}{2}$ and the observed sequence tends to a fair coin: the channel converts structure into noise, with the parity bias 1/(1+p) as the exact carrier.

Theorem II.E (critical loss for objecthood). Let the feature be the flip fraction F over a window of h consecutive flip indicators (and therefore h+1 delivered observations), the alternating type region Balt = $[1-\delta, 1]$ with $\delta \in (0, \frac{1}{2})$, and recognition thresholds drawn from E $\subseteq [0, \varepsilon]$ with s := $\delta + \varepsilon < \frac{1}{2}$. Set

$$pc = s/1 - s, \qquad f = 1/1 + p, \qquad d* = max0, 1 - -f$$

(a) (Dissolution.) If p > pc, then d* > ε and each window is recognized at some threshold t $\leq \varepsilon$ with probability at most $\exp(-2h(d^* - \varepsilon)^2)$. Consequently, for every fixed token length W, the probability of robust recognition over any threshold interval contained in $[0, \varepsilon]$ vanishes exponentially as h → ∞.

(b) (Persistence.) If p < pc, then d* < ε, and for every $0 < \gamma < \varepsilon$ − d* a W-window token is recognized simultaneously at all thresholds t ∈ [d* + γ, ε] with probability at least 1 − W $\exp(-2h\gamma^2)$. Robust segmentation persists with a robustness range of length at least ε − d* − γ.

(c) (Asymmetry.) The frozen type has flip fraction identically 0 and is recognized at every threshold for every p < 1.

Proof. By Lemma II.E1, hF ∼ Bin(h, f). A window is recognized at threshold t iff d(F, Balt) ≤ t, i.e. iff F ≥ 1 − δ − t. (a) For every t ≤ ε this requires F ≥ 1 − s, and p > pc ⟺ f < 1 − s; Hoeffding gives P(F ≥ 1 − s) ≤ $\exp(-2h((1-s) - f)^2)$ with (1−s) − f = d* − ε. (b) p < pc ⟺ f > 1 − s ⟺ d* < ε. Simultaneous recognition on [d* + γ, ε] is implied by recognition at the leftmost threshold, i.e. by F ≥ 1 − δ − d* − γ; whether d* > 0 (where 1 − δ − d* = f) or d* = 0 (where 1 − δ ≤ f), the condition is implied by F ≥ f − γ, and P(F < f − γ) ≤ $\exp(-2h\gamma^2)$; a union bound over W windows completes it, and threshold-monotonicity of the recognized set makes the segmentation constant on the interval; no frozen window is spuriously admitted anywhere on it, since the frozen feature is 0 and its distance to Balt equals 1 − δ > $\frac{1}{2}$ > ε. (c) A constant source delivers a constant symbol under any block parity. □

Corollary II.E1 (fast objects dissolve first). The critical tolerance budget satisfies $\delta + \varepsilon$ = p/(1+p): what must be spent to keep recognizing the fast object equals exactly the channel's parity bias. Since the frozen type survives every loss rate, the segmentation of the pair degrades only through its faster member: in a lossy clock, objecthood is lost in order of intrinsic speed.

Remark (beyond period two). For a period-r source the parity mechanism generalizes: the observed phase performs a random walk on Zr with step law m mod r, m ∼ Geom(q), and recognition statistics are governed by that walk's spectral gap; the critical constant becomes a function of the gap rather than of the parity bias alone. The sharp constant is left to future work.

Numerical check Appendix N.5: the transition is bracketed on both sides of pc = 1/3, with critical slowing at the boundary and monotone convergence in h.

7. Further results (stated, not proved here)

Conjecture F (Markov-modulated sources). If the source is Markov-modulated (increments emitted from hidden states), identifiability of the hidden clock structure from the record is expected to reduce to a Kruskal-rank condition [3] on the emission and transition factors, by an argument analogous to standard tensor-decomposition identifiability proofs. The gauge of Theorem II.C then acts on the identifiable equivalence class rather than on a point.

8. Positioning

What this framework does not claim (i) That time, work, and information are one physical field — they are distinct typed functionals of a shared causal carrier. (ii) That continuous time necessarily emerges from noise — it enters by interpolation, or as a limit of refining clocks. (iii) That Maxwell's demon is hereby dispelled — its thermodynamics is settled by the Szilard–Landauer–Bennett line, on which Part I builds. (iv) That Newtonian time is refuted — only that it is not required as an ontological primitive of this construction. (v) That the question of thinghood is closed — the definition offered is operational and recognition-scheme-relative.

Part I develops the formal architecture — transition, causal order, internal and external clocks, and the delivery channel — while distinguishing that contribution from the established thermodynamic and information-theoretic results it uses. Part II derives quantitative consequences of that architecture. Theorem II.A identifies the exact-constant reconstruction floor and the recoverability dichotomy of Corollary II.A1; the constant is p σ^2 per source tick, turning Part I's "the record is partial" into a rate. Theorem II.C establishes the principal identifiability result: compound-geometric non-identifiability belongs to the established theory of geometric random sums and geometric infinite divisibility [8, 9]. The contribution here is its clock-channel interpretation: an explicit observational-equivalence orbit for the source/channel decomposition, its indecomposable representative, and provenance-based gauge fixing. To the best of our knowledge, that combined operational interpretation has not previously been stated for a delivered physical clock record. The resulting observational-equivalence orbit generated by geometric compounding shows that the source rate is not identifiable independently of the channel. Theorem II.E extends the analysis from time reconstruction to object recognition: the recognition scheme that Part I introduced descriptively acquires a sharp phase boundary, with the critical constant expressed entirely in the scheme's own tolerances.

The sharp critical loss rate established in Theorem II.E is structurally related to the critical observation scale introduced in [12], where model-based regulation becomes effective only above a scale-dependent threshold. The control parameters and observables are different: [12] varies smoothing scale and explained variance, whereas the present result varies channel loss and the robustness of object-token recognition.

Appendix N. Numerical verification

All simulations use NumPy's PCG64 generator; the seeds, sample sizes, and target quantities are reported below to support independent reproduction.

N.1. Theorem II.A

Seed 7. n = 200, p = 0.3, F = Exp(1) ($\mu = \sigma^2 = 1$), $2 \cdot 10^5$ trials: empirical MSE of the conditional-mean decoder 60.18 against the exact floor n p $\sigma^2 = 60.00$ (ratio 1.003). Degenerate calibration $\Delta \equiv 1$: maximal absolute error 0.0 over $2 \cdot 10^4$ trials.

N.2. Lemma II.B1 and Theorem II.B

Seed 11. Lemma II.B1 verified at (n, q) $\in$ {(1000, 0.7), (200, 0.3), (5000, 0.9), (50, 0.5)}: e.g. bound 0.0432 against true maximum 0.0275 at (1000, 0.7). Shift bound TV(Bin(n,q), Bin(n+k,q)) $\leq$ kq$\sqrt{}$(π/(8nqp)) verified at k $\in$ {5, 15, 30}; the prescribed k* = 16, in the (n, q) = (1000, 0.7) regime, yields TV = 0.2996 $\leq \frac{1}{2}$.

N.3. Theorem II.C

Seed 7. $q_1 = 0.4$ versus $q_2 = 0.8$ with $F'' = \mathrm{Geom}(0.5) \circ \mathrm{Exp}(1)$, $4 \cdot 10^5$ samples per side: matching means 2.501 / 2.500 (theory $1/q_1 = 2.5$) and variances 6.245 / 6.253; two-sample KS statistic 0.0037, p-value 0.50 on 10^5-subsamples. Laplace identities at s $\in$ {0.3, 1.0, 2.7}: discrepancy $\leq 1.1 \cdot 10^{-16}$.

N.4. Theorem II.D

Seed 11. Unicyclic three-state register, forward/backward rates 2 : 1, with N defined as the net winding current, T = 300, 4000 Gillespie trials: $\mathrm{Var(N)}/\langle \mathrm{N} \rangle^2 = 0.00998$ against $2/\Sigma = 0.00962$ (within 4% of TUR saturation); with w $\sim$ Exp(1) the full chain realizes as $0.01356 \geq 0.00998 \geq 0.00962$.

N.5. Lemma II.E1 and Theorem II.E

Seeds 5, 9. Flip frequencies match 1/(1+p) to four digits at p $\in$ {0.2, 0.5, 0.8}, lag-1 correlations below 10^{-3}. With $\delta = 0.10$, $\varepsilon = 0.15$ (pc = 1/3), margins $\gamma = (\varepsilon - \mathrm{d}^*)/2$, W = 20 windows: robust-recognition probability 1.000 at p = 0.15 (h = 800), 0.952 at p = 0.25 (h = 2000), 0.657 at p = 0.30 (h = 8000; critical slowing), identically 0 at p $\in$ {0.35, 0.45, 0.60}, where the admissible interval has negative length. Convergence in h at p = 0.25: 0.029, 0.195, 0.619, 0.949, 0.999 at h = 200, 500, 1000, 2000, 4000.

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ONTOLOGY OF TRANSITION—PART III

The Thermodynamic Price of External Time:
Rate–Distortion Bounds for Physical Clock Records

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Abstract

This article is Part III of the three-part scientific work *Ontology of Transition.* The work's central idea is that transition, rather than the static thing, is the primary unit of description, while operational time is realized through physical records of transitions whose production, transmission, interpretation, and reuse have distinct informational and thermodynamic limits.

External time is operationally available to a system only through a physical record of a selected clock process, delivered through a possibly noisy, incomplete channel KCS, (Part I [1]). Part II [2] quantified two costs of that mediation: a channel information floor on reconstruction (its Theorem II.A) and a thermodynamic precision floor on the reading register (its Theorem II.D). This paper adds a renewal-work bound to the series' cost decomposition. For a finite classical footprint with equal-free-energy logical states, exact isothermal renewal without accessible correlated side information obeys $\beta\langle$Wrenew$\rangle$ $\geq$ RKclock(D) $\geq$ R(D), where RKclock is the indirect (remote) rate–distortion function determined by the physical clock-delivery channel — the classical indirect rate–distortion problem induced by the delivered clock record [3, 4, 5]. The bound extends to correlated finite histories without i.i.d. assumptions; closed forms are derived for a uniform M-phase clock behind a symmetric channel and behind an erasure-and-substitution channel, with the exact indirect frontier proved via the Wolf–Ziv reduction; a sequential-ring corollary shows the asymptotic renewal cost equals the entropy of innovations, not the tick count. Target distortions below the channel's Bayes risk are unattainable at any reset budget. The three floors — information destroyed in the channel, precision paid in register dissipation, reuse paid in reset work — constrain different observables by different mechanisms and none implies another. The result does not identify time with work; it prices one specified operation: making a prescribed resolution of external time reusable at a chosen system boundary. All closed forms are verified against a Blahut–Arimoto solver to machine precision, and the literal decoder attains the indirect frontier at minimum distortion exactly. The causal (nonanticipative) version is posed as the open target.

Keywords: *external time; rate–distortion; Landauer principle; clock records; renewal work; indirect source coding*

1. From ontology to reset work

Part I [1] establishes that the external time accessible to a system is not a direct reading of the metasystem: it is a functional of a physical record produced by a selected clock process and delivered through a possibly noisy, incomplete, and history-dependent channel KCS„ and it separates the costs of signal generation, transmission, registration, storage, and memory reset. Part II [2] made two of those costs quantitative: the reconstruction error that the channel itself destroys (an exact minimax floor of n p σ^2 per window, [2, Thm. A]) and the precision that the reading register must pay for in dissipation while it runs (a thermodynamic uncertainty floor 2/reg, [2, Thm. D]). What remained unpriced is the cost that appears only when the observer is reusable: the record and everything computed from it must eventually be returned to a standard state.

This paper proves the following bound:

> **A reusable observer cannot reconstruct an external clock to prescribed accuracy and then complete a closed renewal of the entire information-bearing footprint for less work than the rate–distortion information required by that reconstruction.**

In the symmetric Landauer regime, and under the assumptions of Theorem III.1 below,

$$\langle Wrenew\rangle kBT\ RKclockDkBT\ RD,$$

where D is the allowed clock-reconstruction distortion, RKclock(D) is the indirect (remote) rate–distortion function fixed by the actual clock-delivery channel, and R(D) is the ordinary rate–distortion function of the hidden source reading. This is not the identity "time is work." It is a conditional lower bound on one specified physical operation: resetting a reusable record of external time.

2. Operational model

Let $\Theta \in$ T be the hidden reading of a selected external clock process; $\rho \in$ Y the record delivered through KCS,; G $\in$ G the complete finite information-bearing footprint to be returned to its standard state, including the raw register, decoded output, correlated workspace, and any copies that do not survive renewal; $\widehat{\Theta}$ the decoded clock reading; d(θ, θ) ≥ 0 a specified distortion function. The information flow is $\Theta \to \rho \to$ G $\to \widehat{\Theta}$, and the target accuracy is E d(Θ, $\widehat{\Theta}$) $\leq$ D. All entropies and mutual informations use natural logarithms (nats).

(Notation guard: throughout the series, p denotes the loss probability of the clock channel and q = 1 − p its delivery or flush probability [2, §6]. In this paper the record channel's substitution error is therefore written u and its erasure probability ε; q keeps its series meaning and is not reused.)

Define the ordinary source rate–distortion function

$$RD := infP| : EdDI; ,$$

and, for the fixed physical delivery law P(θ, ρ) = P(θ) KCS,($\rho|\theta$), the indirect clock rate–distortion function

$$RKclockD := infP| : EdDI;$$

The second quantity is the minimum information rate that must be retained from the actually delivered record, rather than from an unrealistically available source state. If the requested distortion is below the Bayes risk imposed by the channel, the feasible set is empty and RKclock(D) = $+\infty$ by convention.

3. The classical reduction

The indirect rate–distortion function is a classical object: introduced by Dobrushin and Tsybakov [3], given its operational transmission form by Wolf and Ziv [4], and named and systematized by Witsenhausen [5]. Its key structural property, which this paper uses twice, is the reduction to an ordinary problem on the record alphabet.

Lemma III.1 (Wolf–Ziv reduction [4]). Define the modified distortion d(ρ, θ) := E[d(Θ, θ) | ρ]. Then

$$RKclockD = infP \,|: \, Ed, DI; ,$$

i.e. the indirect problem for Θ observed through K equals the ordinary rate–distortion problem for the source ρ under d.

Proof. For any decoder P($\theta|\rho$), the Markov chain $\Theta \to \rho \to \widehat{\Theta}$ gives E d(Θ, $\widehat{\Theta}$) = E[E[d(Θ, $\widehat{\Theta}$) | ρ, $\widehat{\Theta}$]] = E[d(ρ, $\widehat{\Theta}$)], since conditionally on ρ the pair (Θ, $\widehat{\Theta}$) is independent. The two constraint sets therefore coincide, and the objective I(ρ; $\widehat{\Theta}$) is the same. □

The primitive inequalities of the next section are likewise classical — Shannon's rate–distortion theory [6], Landauer's bound [7], the information thermodynamics of measurement and erasure

[8] — and are not claimed as new. The contribution is the composite bound and its clock-record interpretation (§13).

4. Clock-record rate–distortion–renewal theorem

Theorem III.1 (single record). Assume that: (1) G is a finite classical memory whose logical states have equal internal free energy at the beginning and end of the renewal protocol; (2) it is reset exactly to one standard state while coupled to a bath at temperature T; (3) the reset controller has no access to Θ, ρ, a surviving copy of $\widehat{\Theta}$, a correlated random seed, or any other side information with which the record could be uncomputed; (4) G includes the entire clock-correlated footprint that is erased; in particular, the decoded output cannot be retained elsewhere and silently excluded from the accounting; (5) Wrenew is the average work supplied to return G to its standard state, not the dissipated-work quantity W − ΔF.

The completeness requirement on G is necessarily boundary-relative. Determining which registers, copies, correlated workspaces, and surviving records belong to the renewed system is not merely an implementation detail: it specifies the physical boundary across which information and work are accounted. A related formal treatment of the system boundary as an explicit information-bearing object of control is developed in [18]. In the present theorem, the boundary is fixed rather than optimized, but it must be stated explicitly for the reset-work attribution to be well posed.

Under these assumptions, every implementation satisfying E d(Θ, $\widehat{\Theta}$) ≤ D obeys

$$\langle Wrenew\rangle HGI; GI; RKclockDRD, = kBT - 1$$

Proof. The symmetric Landauer bound for exact local reset gives $\beta\langle$Wrenew$\rangle \geq$ H(G). Trivially H(G) ≥ I(ρ; G). Because $\rho \to$ G $\to \widehat{\Theta}$, the data-processing inequality gives I(ρ; G) ≥ I(ρ; $\widehat{\Theta}$). The induced decoder P($\theta|\rho$) is one of the kernels over which RKclock(D) is minimized, hence I(ρ; $\widehat{\Theta}$) ≥ RKclock(D). Finally, $\Theta \to \rho \to \widehat{\Theta}$ implies I(ρ; $\widehat{\Theta}$) = I(Θ; $\widehat{\Theta}$) + I(ρ; $\widehat{\Theta}$ | Θ) ≥ I(Θ; $\widehat{\Theta}$) ≥ R(D); taking the infimum over admissible record decoders yields RKclock(D) ≥ R(D). □

What the two lower bounds mean kB T R(D) is the best possible floor for an observer that could encode the source reading directly. kB T RKclock(D) includes the informational penalty of receiving only the partial record ρ. The gap RKclock(D) − R(D) ≥ 0 is a precise candidate for the price of mediated access to external time.

5. Finite-history version

Let Θ^n be a hidden clock history, ρ^n its delivered record, Gn the complete block footprint renewed after use, and $\widehat{\Theta}^n$ the reconstruction, with the per-symbol distortion dn = (1/n) Σ_i d(θ_i, θ_i). Define RK,nclock(D) := (1/n) inf over P($\theta^n|\rho^n$) with E dn ≤ D of I(ρ^n; $\widehat{\Theta}^n$), and the direct block function $R^{n(n)}$(D) analogously. The same proof, with vectors replacing single variables, gives

$$\langle Wrenewn\rangle nRK, nclockDRnnD$$

The finite-block inequality accommodates correlated clock histories and a history-dependent channel KCS,; no i.i.d. assumption is needed. Stationarity or ergodicity becomes relevant only in the long-history limit.

6. Closed form: a uniform M-phase clock

Let Θ be uniform on ZM with Hamming distortion d(θ, θ) = 1{$\theta \neq \theta$}. If the clock-reading error probability is pe, then for 0 ≤ pe ≤ 1 − 1/M the ordinary rate–distortion function is R(pe) = ln M − h_2(pe) − pe ln(M−1), where h_2 is the binary entropy (nats). Hence ⟨Wrenew⟩ ≥ kB T [ln M − h_2(pe) − pe ln(M−1)]; for exact reconstruction ⟨Wrenew⟩ ≥ kB T ln M; for pe ≥ 1 − 1/M a constant guess meets the target and the floor is zero.

Channel-aware closed form Let the record pass through an M-ary symmetric channel with substitution error 0 ≤ u < 1 − 1/M. The posterior given ρ assigns probability 1 − u to Θ = ρ and u/(M−1) to each other value, so Lemma III.1's modified distortion is affine in the Hamming indicator on the record alphabet:

$$d, = u + c\,1, c = 1 - Mu/M - 1 > 0$$

The constraint E d ≤ D is therefore equivalent to a Hamming constraint on (ρ, $\widehat{\Theta}$) at level δ(D, u) = (D − u)/c, the feasibility floor is D ≥ u (the Bayes risk), and since ρ is itself uniform on ZM, Lemma III.1 reduces the indirect problem to the ordinary uniform-source problem at level δ:

$$RKclockD = +forD < u; \quad lnM - h_2 - lnM - 1foruD1 - 1/M; \quad 0forD1 - 1/M,$$

continuously, since δ(1 − 1/M, u) = 1 − 1/M. Channel noise does not merely change the achieved error: it raises the minimum reusable-memory rate, and target distortions below u are physically unattainable regardless of reset work.

7. Minimal test model: loss, substitution, and raw-memory overhead

Take a uniform M-phase clock and a delivered record for which the complete renewed footprint is just the raw register, G = A = ρ ∈ ZM ∪ {⊥}. For every true phase θ: an erasure ⊥ occurs with probability ε; conditional on non-erasure, the correct phase is delivered with probability 1 − u and each wrong phase with probability u/(M−1). Decoding a received phase literally and guessing uniformly after erasure gives D = (1−ε)u + ε(1 − 1/M). The raw register entropy is H(A) = h_2(ε) + (1−ε) ln M, and the clock information reaching it is I(Θ; A) = (1−ε)[ln M − h_2(u) − u ln(M−1)]. The induced reconstruction channel Θ → $\widehat{\Theta}$ is M-ary symmetric at error D, so I(Θ; $\widehat{\Theta}$) = R(D), and the model separates the quantities through two valid chains: H(A) ≥ I(A; $\widehat{\Theta}$) ≥ RKclock(D) ≥ R(D), and H(A) ≥ I(Θ; A) ≥ I(Θ; $\widehat{\Theta}$) = R(D). There is no universal ordering between I(A; $\widehat{\Theta}$) and I(Θ; A); they answer different questions.

Proposition III.1 (exact indirect frontier). Put DB = (1−ε)u + ε(1 − 1/M), c = 1 − Mu/(M−1), and δ(D) = (D − DB)/((1−ε)c) for D ∈ [DB, 1 − 1/M]. Then

$$RKclockD = 1 - lnM - h_2 - lnM - 1,$$

with RKclock = +∞ for D < DB and 0 for D ≥ 1 − 1/M. In particular RKclock(DB) = (1−ε) ln M.

Proof. By Lemma III.1 compute d on the record alphabet. For a = ⊥ the posterior of Θ is uniform, so d(⊥, θ) = 1 − 1/M for every θ: the erased branch contributes the constant ε(1 − 1/M) to distortion for any decoder and can contribute zero rate. For a ≠ ⊥, as in §6, d(a, θ) = u + c · 1{θ ≠ a}. Writing B = 1{ρ = ⊥} (a function of ρ) and π = P($\widehat{\Theta}$ ≠ ρ | B = 0), the constraint E d ≤ D is equivalent to $\pi \le \delta(D)$. Converse: $I(\rho; \widehat{\Theta}) = I(B; \widehat{\Theta}) + I(\rho; \widehat{\Theta} \mid B) \ge (1-\varepsilon)\, I(\rho; \widehat{\Theta} \mid B = 0) \ge (1-\varepsilon)[\ln M - h_2(\pi) - \pi \ln(M-1)] \ge (1-\varepsilon)[\ln M - h_2(\delta) - \delta \ln(M-1)]$, using that $\rho \mid B = 0$ is uniform on ZM and that the uniform-source Hamming rate–distortion function is nonincreasing. Achievability: on B = 0 use the optimal symmetric test channel at Hamming level δ (whose output marginal is uniform); on B = 1 draw $\widehat{\Theta}$ uniformly, independently of everything. Then $\widehat{\Theta}$ is uniform and independent of B, so $I(B; \widehat{\Theta}) = 0$ and $I(\rho; \widehat{\Theta} \mid B = 1) = 0$, giving $I(\rho; \widehat{\Theta}) = (1-\varepsilon)[\ln M - h_2(\delta) - \delta \ln(M-1)]$ exactly. □

Remark (exact attainment). The frontier of Proposition III.1 is attained at DB by the literal decoder itself: its output marginal is uniform on ZM ($H(\widehat{\Theta}) = \ln M$), its conditional entropy is $H(\widehat{\Theta} \mid A) = \varepsilon \ln M$ (deterministic on delivery, uniform on erasure), hence $I(A; \widehat{\Theta}) = (1-\varepsilon)$ ln M = RKclock(DB), while its distortion equals DB by construction. The minimum-distortion point of the indirect frontier is therefore achieved exactly, not merely approached.

Consequently H(A) − RKclock(DB) = $h_2(\varepsilon)$ is the raw cost of retaining the erasure flag, while RKclock(DB) − R(DB) quantifies the additional rate required because the observer encodes a noisy record rather than the hidden clock state itself.

Numerical checkpoint For M = 8, ε = 0.2, u = 0.1: D = 0.255; H(A) = 2.164 nat, I(Θ; A) = 1.248 nat, RKclock(DB) = 1.664 nat, R(D) = 1.015 nat. At T = 300 K (kB T ≈ 4.142 zJ): ideal reset of the raw register requires at least ≈ 8.96 zJ; the direct source floor is ≈ 4.21 zJ; the stronger channel-aware floor at minimum attainable distortion is ≈ 6.89 zJ. The differences are the energetic prices of raw representation, channel constraints, and unused record entropy respectively. Appendix N verifies both closed forms against a Blahut–Arimoto solver to machine precision; the exact attainment at DB is the Remark above.

Figure III.1. Thermodynamic rate–distortion hierarchy for the noisy external-clock record at (M, ε, u) = (8, 0.2, 0.1): raw-register cost H(A), channel-aware frontier R_K^clock(D), source–record information I(Θ; A), and direct frontier R_Θ(D) separate as the allowed distortion varies; the literal decoder sits on the frontier at D_B. Curves are the solver-verified closed forms of §6 and Proposition III.1.

8. Sequential ring corollary: renewal pays for innovation, not ticks

Consider a genuinely evolving M-phase clock k+1 = k + 1 + Jk (mod M), with Θ_0 uniform and independent jitter P(Jk = 0) = 1 − η, P(Jk = ±1) = $\eta/2$. For M ≥ 3 the step alphabet {0, 1, 2} is injective mod M, so the increments are recoverable from the recorded path, and an exact length-n history has entropy $H(\Theta_0, \ldots, n) = \ln M + n[h_2(\eta) + \eta \ln 2]$. Lossless closed renewal therefore obeys

$$\langle Wrenewn\rangle/nh_2 + ln2 + lnM/n$$

In the long-history limit the work floor per step is the entropy of the unpredictable innovations, not the number of ticks. For a deterministic ring ($\eta = 0$) only the initial phase must be retained: the optimally compressed renewal cost per tick tends to zero even though the clock keeps changing state. This is a quantitative corollary of the separation principle of [1].

9. Tightness

The inequality is one-shot, but simultaneous equality is generally unavailable for a single noisy symbol: H(G) = I(Θ; G) requires G conditionally deterministic given Θ, whereas a single-letter rate–distortion optimum is often stochastic. The bound becomes operationally tight in the asymptotic i.i.d. setting if long record blocks are encoded by a deterministic rate–distortion code; the code index is the only clock-correlated memory that remains; encoding, reconstruction, and workspace cleanup are performed reversibly or fully included in the accounting; and the code-index register is reset quasistatically. Then H(Gn)/n $\to$ RKclock(D) and $\langle$Wrenew$^{(n)}\rangle$/n $\to$ kB T RKclock(D). Finite-time reset, finite baths, imperfect control, and uncompressed raw records all increase the required work.

10. Three floors of the accessible clock

The series now prices the accessible clock coordinate three times, and the three floors are mathematically disjoint.

Floor 1 — channel information ([2], Theorem II.A) Reconstruction: the minimax mean-squared error through the erasure channel is exactly n p σ^2 per window. It constrains an estimation error, is enforced by conditioning (the exact Bayes filter), and is paid in lost calibration fluctuation. It binds even for a dissipationless, never-reset observer.

Floor 2 — register precision ([2], Theorem II.D) Reading: for a register in a nonequilibrium steady state, Var(t)/$\langle$t$\rangle^2 \geq$ 2/reg. It constrains the relative variance of a current, is enforced by the thermodynamic uncertainty relation, and is paid in the register's own dissipation while it runs. It binds even for a perfect channel (σ^2 = 0, p = 0), where Floor 1 vanishes.

Floor 3 — renewal work (Theorem III.1 here) Reuse: $\beta\langle$Wrenew$\rangle \geq$ RKclock(D). It constrains the average reset work, is enforced by Landauer's bound plus data processing, and is chargeable even at zero entropy production, since a quasistatic reset saturates it: it bounds work, not dissipation. It binds even for an infinitely dissipative register and vanishes only if the observer never reuses its footprint.

No floor implies another: Floor 1 survives reg $\to \infty$ and W $\to \infty$; Floor 2 survives σ^2 = 0; Floor 3 survives reg $\to$ 0. They constrain three different observables (mean-squared error; relative variance; work) by three different mechanisms (Bayes conditioning; TUR; Landauer). The channel is nonetheless present inside Floor 3: RKclock internalizes it, and its domain boundary — infeasibility below the Bayes risk — is Floor 1's destruction re-expressed as unattainability. Two scope remarks carry over. Floor 2 is proved in [2] for erasure-type registers and fails for the aggregating channel by weight–count anticorrelation. Floor 1's underlying Bayes-variance form, risk $\geq$ E[Var(tC | record)], is channel-agnostic, but its exact constant n p σ^2 is specific to the erasure channel of [2]; Floor 3 is channel-agnostic outright — any delivery law enters only through RKclock.

11. What the theorem does not say

The theorem does not establish a universal energetic cost of time. It does not imply that: every clock tick dissipates kB T times some fixed information; measurement or discrimination necessarily has the Landauer cost at the moment it occurs; generation, transmission, registration, storage, and reset have one interchangeable cost; time, work, information, or entropy production are identical; a memory that is never reset has already paid the reset bound; or that correlations may be used during reset for free. The reset-work bound can be saturated quasistatically with zero irreversible entropy production, and therefore must not be restated as a lower bound on dissipated work or on Σ.

If correlated side information Z survives and is available to the reset controller, then for any decoder of the form = δ(G, Z) the conditional extension is $\beta\langle$Wrenew$\rangle \geq$ H(G | Z) $\geq$ I(Θ; G | Z) $\geq$ I(Θ; | Z) $\geq$ R|Z(D), and the local cost can be smaller [9]; if Z survives but is unavailable to the controller, the unconditional H(G) is the relevant cost. Information about external time may migrate between registers, but deleting the last usable copy is the operation to which the closed-renewal cost belongs. Accordingly, the system boundary must identify both the information-bearing footprint being renewed and every surviving copy excluded from that renewal accounting [18]. The source also needs a nontrivial prior — a random query time, a stochastic clock path, or uncertainty induced by partial observation; if Θ is deterministic its rate–distortion function is zero and the bound is vacuous. For nondegenerate memory states the first step must be replaced by the appropriate generalized free-energy inequality; the simple kB T H(G) form is specific to the symmetric equal-free-energy model of [1].

12. The open target: causal external-time rate–distortion

The finite-history theorem permits block encoding after the complete record is available; a real observer reconstructs time online. A causal extension would impose nonanticipation, P(θ_i | ρ^n, θ^{i−1}) = P(θ_i | ρ^i, θ^{i−1}), leading to a causal indirect rate–distortion function, naturally expressed through directed information, with target statement liminf $\beta\langle$Wrenew$^{(n)}\rangle$/n $\geq$ RKcausal(D). That extension would exercise the full distinctive architecture of [1]: causal partial order, a selected clock subsystem, loss, delay, duplication, aggregation and out-of-order delivery, finite reusable memory, and physically implemented reset. The single-record theorem is the present foundation; a causal history theorem remains open.

13. Relation to prior work

The proof ingredients are established: Shannon rate–distortion theory [6]; the indirect rate–distortion function itself [3, 4, 5]; Landauer erasure [7]; data processing; information thermodynamics of measurement and erasure [8]; model-specific accuracy–dissipation relations for physical clocks [10, 11, 12]. None of the primitive inequalities is claimed as new. The specific contribution is their integration into the external-clock architecture of [1, 2]: the hidden clock state belongs to a metasystem outside the observed boundary; the observer has access only to a record delivered through KCS,; clock accuracy is a reconstruction distortion; reusable memory carries an explicit reset cost; and the resulting bound separates direct source complexity, channel-mediated complexity, and raw-register overhead. The explicit treatment of that boundary as part of the information and work attribution is aligned with the boundary formalism developed in [18]. To the best of our knowledge, this is the first explicit rate–distortion lower bound on the thermodynamic renewal cost of a reusable, lossy external-time record.

Closest precedents and the remaining gap Sagawa & Ueda [8]: work bounds for measurement and erasure in terms of entropy and mutual information — no external clock, reconstruction distortion, or clock-delivery channel. Erker et al. [11]: autonomous clockwork with a tick register whose reset is recognized as an additional Landauer cost — the derived accuracy–dissipation relation excludes record measurement/erasure; no lossy decoder or rate–distortion bound. Still, Sivak, Bell & Crooks [13]: dissipation bounded by non-predictive information retained in a driven system's memory — prediction of a driving signal, not renewal of an external-time record through a specified channel. Gagliardi, Pecchia & Di Carlo [14]: rate–distortion in thermodynamic feedback — feedback benefit, not renewal cost. Nair [15]: rate–distortion limits in optical metrology — probe/channel energy, not Landauer reset work of reusable memory. Gammaitoni [16]: imperfect binary erasure of the form kB T[ln 2 − h_2(p)] — the error belongs to the erasure operation itself, not to a clock record subsequently reset exactly. del Rio et al. [9]: side information turns erasure work into a conditional-entropy problem — supplies the boundary condition of §11, not the clock theorem. Hsieh [17]: bounds information transmission by work extractable from maintained correlations — distinct from the cost of erasing a reusable channel record.

14. Outlook

(1) Extend the Blahut–Arimoto verification of Appendix N to record alphabets with delay and out-of-order delivery. (2) Prove the nonanticipative finite-memory version using directed information. (3) Derive conditional bounds when the observer retains side information. (4) Connect the information floor to a concrete reset protocol for a double-well, nanomagnetic, or electronic memory [12]. (5) Separate measured work into generation, transmission, registration, compression, and reset terms, closing the loop with the cost taxonomy of [1, §11]. The division across the three parts is therefore explicit: Part I says external time is a functional of a partial physical record; Part II says what the channel and the running register irreducibly cost; Part III says how much reusable physical memory — and therefore how much ideal reset work — is required to realize that functional at a chosen accuracy.

Appendix N. Numerical verification

All computations use NumPy; the stochastic checks report their PCG64 seeds, sample sizes, and target quantities to support independent reproduction.

N.1. Channel-aware closed form (§6)

A Blahut–Arimoto solver (fixed-point iteration on the output marginal, tolerance 10^{-13}, with bisection on the Lagrange slope to hit each target distortion) was run on the record alphabet ZM with the modified distortion of Lemma III.1. Twelve points across four regimes (M, u) $\in$ {(8, 0.10), (5, 0.20), (3, 0.05), (12, 0.15)}, at 15%, 50%, and 85% of the feasible distortion range: the solver agrees with the closed form at every point, worst deviation $8.9 \cdot 10^{-16}$ nat — machine precision. The solver has no knowledge of the closed form.

N.2. Exact indirect frontier (Proposition III.1)

The same solver on the (M+1)-letter alphabet ZM $\cup$ {$\perp$} with d($\perp$, $\cdot$) = 1 − 1/M constant: nine points across (M, ε, u) $\in$ {(8, 0.2, 0.1), (5, 0.3, 0.15), (6, 0.1, 0.05)} at 20%, 55%, and 90% of the feasible range agree with Proposition III.1 at every point, worst deviation $5.0 \cdot 10^{-16}$ nat.

N.3. Ring history entropy (§8)

Exact enumeration of all path laws at (M, n, η) = (3, 6, 0.3) and (5, 4, 0.2): the map (Θ_0, J_1,. . . ,Jn) $\mapsto$ (Θ_0,. . . ,n) is bijective ($2187 = 3 \cdot 3^6$ and $405 = 5 \cdot 3^4$ distinct paths respectively — the M = 3 boundary of the hypothesis is exercised), and the exact path entropy equals ln M + n[$h_2(\eta) + \eta \ln 2$] to $4 \cdot 10^{-14}$.

N.4. Test model: end-to-end chain

Seed 13, 10^6 samples at (M, ε, u) = (8, 0.2, 0.1): empirical D = 0.2545 (theory 0.2550), H(A) = 2.1642 (2.1640), I(Θ; A) = 1.2490 (1.2478), I(Θ; $\widehat{\Theta}$) = 1.0168 (R(D) = 1.0155). The literal decoder gives $I(A; \widehat{\Theta}) = 1.6643$ against $(1-\varepsilon) \ln M = 1.6636$, confirming the exact attainment of the Remark after Proposition III.1 within sampling error. The full chain realizes as $2.164 \geq 1.664 \geq 1.664 \geq 1.015$. Lemma III.1's identity $\mathrm{E}\, d(\Theta, \widehat{\Theta}) = \mathrm{E}\, d(\rho, \widehat{\Theta})$ was additionally verified on a randomly drawn decoder kernel: 0.87096 vs 0.87100 (sampling error $3.5 \cdot 10^{-5}$).

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